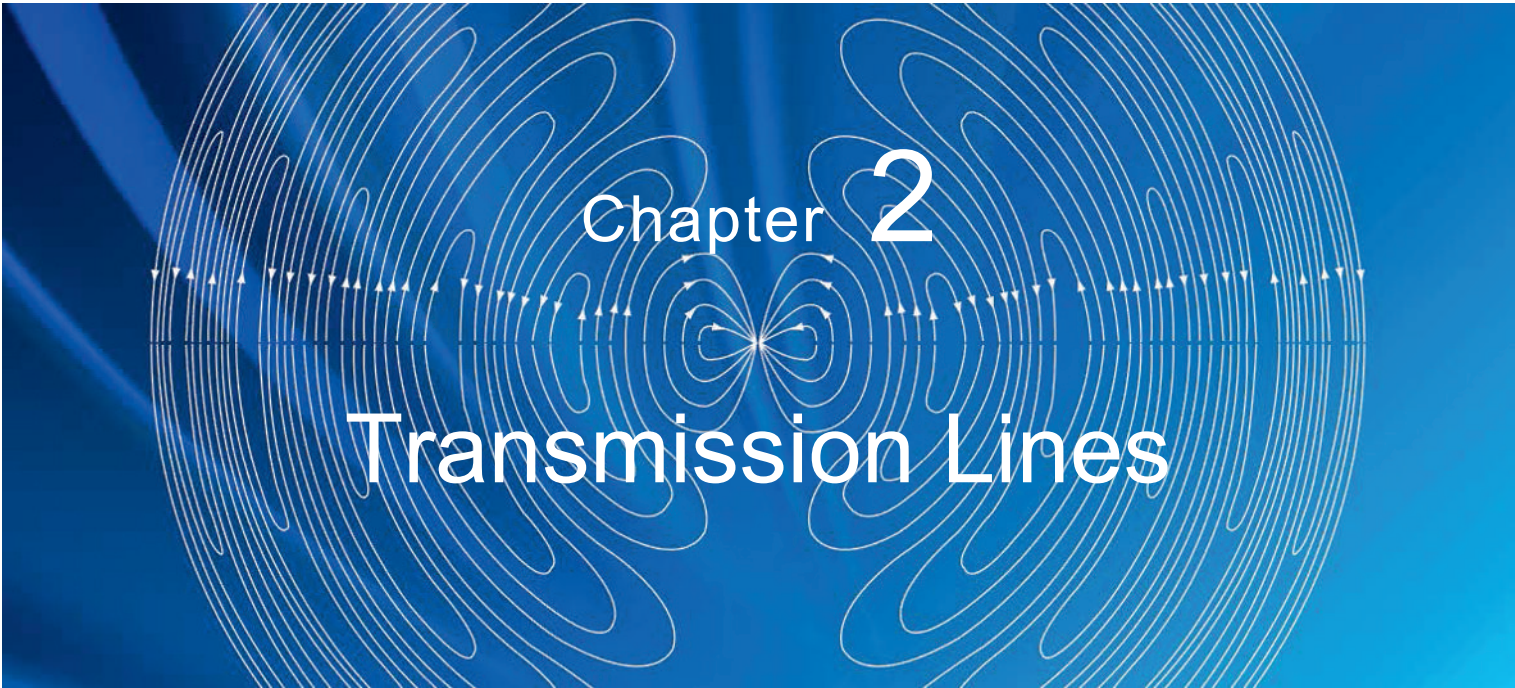




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Objectives

- Upon learning the material presented in this chapter, you should be able to:
- 1. Calculate the line parameters, characteristic impedance, and propagation constant of coaxial, two-wire, parallel-plate, and microstrip transmission lines.
 - 2. Determine the reflection coefficient at the load-end of the transmission line, the standing-wave pattern, and the locations of voltage and current maxima and minima.
 - 3. Calculate the amount of power transferred from the generator to the load through the transmission line.
 - 4. Use the Smith chart to perform transmission-line calculations.
 - 5. Analyze the response of a transmission line to a voltage pulse.

章节内容

2-1 一般注意事项, 47 2-2 集总元件模型, 50 2-3 传输线方程, 53 2-4 传输线上的波传播, 54 2-5 无损微带线, 59 2-6 无损传输线: 一般注意事项, 62 2-7 无损线路的波阻抗, 71 2-8 传输线的特殊情况无损线路, 74 TB3 微波炉, 80 2-9 无损传输线路上的功率流, 82 2-10 史密斯圆图, 84 2-11 阻抗匹配, 94 2-12 传输线路上的瞬态, 108 TB4 EM 癌症消灭器, 117 第 2 章摘要, 119 问题, 121

目标

- 学习本章介绍的材料后, 您应该能够:
- 1. 计算同轴、两线、平行板和微带传输线的线路参数、特性阻抗和传播常数。
 - 2. 确定传输线负载端的反射系数、驻波模式以及电压和电流最大值和最小值的位置。
 - 3. 计算通过传输线从发电机传输到负载的电量。
 - 4. 使用史密斯圆图执行传输线计算。
 - 5. 分析传输线对电压脉冲的响应。

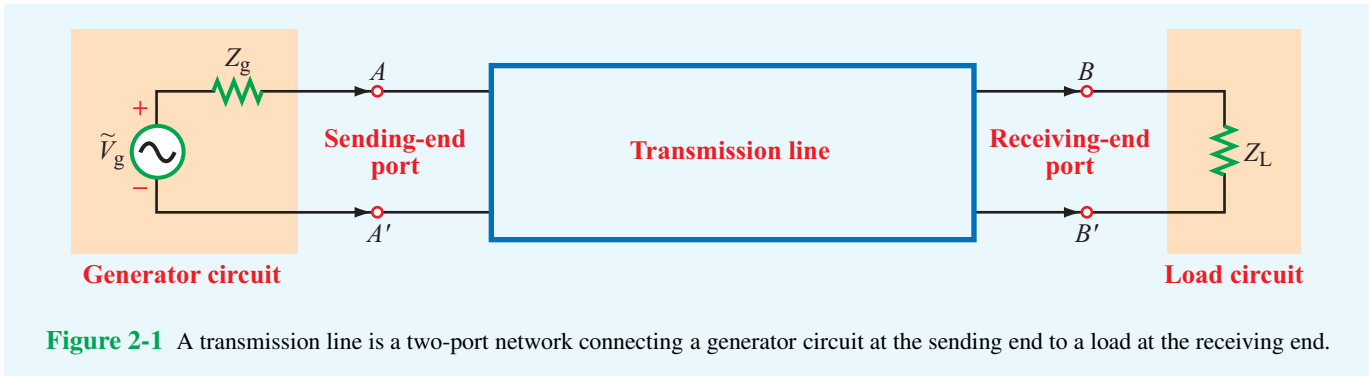


Figure 2-1 A transmission line is a two-port network connecting a generator circuit at the sending end to a load at the receiving end.

2-1 General Considerations

In most electrical engineering curricula, the study of electromagnetics is preceded by one or more courses on electrical circuits. In this book, we use this background to build a bridge between circuit theory and electromagnetic theory. The bridge is provided by transmission lines, which is the topic of this chapter. By modeling transmission lines in the form of equivalent circuits, we can use Kirchhoff's voltage and current laws to develop wave equations whose solutions provide an understanding of wave propagation, standing waves, and power transfer. Familiarity with these concepts facilitates the presentation of material in later chapters.

Although the notion of **transmission lines** may encompass all structures and media that serve to transfer energy or information between two points, including nerve fibers in the human body and fluids and solids that support the propagation of mechanical pressure waves, this chapter focuses on transmission lines that guide electromagnetic signals. Such transmission lines include telephone wires, coaxial cables carrying audio and video information to TV sets or digital data to computer monitors, microstrips printed on microwave circuit boards, and optical fibers carrying light waves for the transmission of data at very high rates.

Fundamentally, a transmission line is a two-port network, with each port consisting of two terminals, as illustrated in Fig. 2-1. One of the ports, the line's sending end, is connected to a source (also called the generator). The other port, the line's receiving end, is connected to a load. The source connected to the transmission line's sending end may be any circuit generating an output voltage, such as a radar transmitter, an amplifier, or a computer terminal operating in transmission mode. From circuit theory, a dc source can be represented by a Thévenin-equivalent **generator circuit** consisting of a generator voltage V_g in series with a generator resistance R_g , as shown in Fig. 2-1. In the case of alternating-current (ac) signals, the

generator circuit is represented by a voltage phasor $\tilde{V}_g$ and an impedance Z_g .

The load circuit, or simply the **load**, may be a transmitting antenna in the case of radar, a computer terminal operating in the receiving mode, the input terminals of an amplifier, or any output circuit whose input terminals can be represented by an equivalent load impedance Z_L .

2-1.1 The Role of Wavelength

In low-frequency circuits, circuit elements usually are interconnected using simple wires. In the circuit shown in Fig. 2-2, for example, the generator is connected to a simple RC load via a pair of wires. In view of our definition in the preceding paragraphs of what constitutes a transmission line, we pose the question: *Is the pair of wires between terminals AA' and terminals BB' a transmission line?* If so, under what set of circumstances should we explicitly treat the pair of wires as a transmission line as opposed to ignoring their presence altogether and treating the circuit as only an RC load connected to a generator $\tilde{V}_g$? The answer to the first question is: *Yes*,

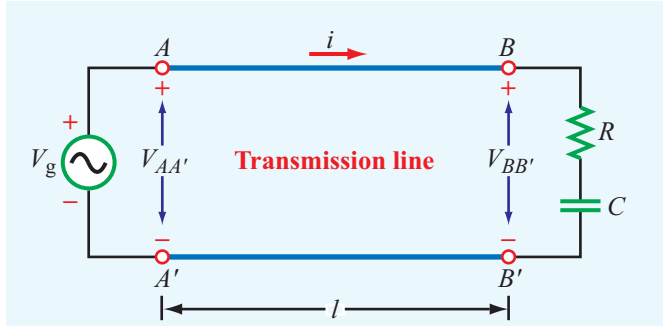


Figure 2-2 Generator connected to an RC circuit through a transmission line of length l .

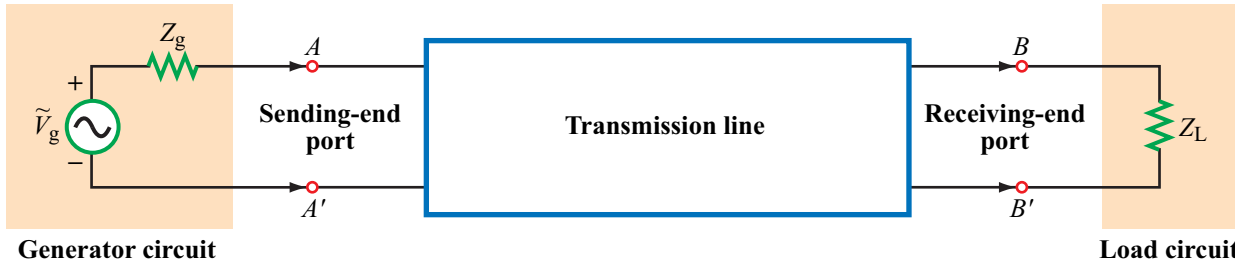


图2-1 A 传输线是连接发送端发生器电路的二端口网络，接收端连接到接收端的负载。

2-1 一般注意事项

在大多数电气工程课程中，电磁学学习之前都会有一门或多门电路课程。在本书中，我们利用这一背景搭建了电路理论和电磁理论之间的桥梁。电桥由传输线提供，这是本章的主题。通过以等效电路的形式对传输线进行建模，我们可以使用基尔霍夫电压和电流定律来建立波动方程，其解可以帮助我们理解波传播、驻波和功率传输。熟悉这些概念有助于在后面的章节中介绍材料。

尽管传输线的概念可能涵盖用于在两点之间传输能量或信息的所有结构和介质，包括人体中的神经纤维以及支持机械压力波传播的流体和固体，但本章重点讨论引导电磁信号的传输线。此类传输线包括电话线、将音频和视频信息传送到电视机或将数字数据传送到计算机显示器的同轴电缆、印在微波电路板上的微带以及传送用于以极高速率传输数据的光波的光纤。

从根本上来说，传输线是一个双端口网络，每个端口由两个终端组成，如图2-1所示。其中一个端口（线路的发送端）连接到源（也称为发生器）。另一个端口，即线路的接收端，连接到负载。连接到传输线发送端的源可以是任何产生输出电压的电路，例如雷达发射器、放大器或在传输模式下运行的计算机终端。从电路理论来看，直流电源可以用 Thévenin 等效发电机电路来表示，该电路由发电机电压 V_g 与发电机电阻 R_g 串联组成，如图 2-1 所示。对于交流 (ac) 信号，

发生器电路由电压相量 $\tilde{V}_g$ 和阻抗 Z_g 表示。

负载电路，或者简称为负载，可以是雷达的发射天线、在接收模式下运行的计算机终端、放大器的输入端子，或者其输入端子可以用等效负载阻抗 Z_L 表示的任何输出电路。

2-1.1 波长的作用

在低频电路中，电路元件通常使用简单的电线互连。例如，在图2-2所示的电路中，发电机通过一对电线连接到简单的RC负载。鉴于我们在前面几段中对传输线构成的定义，我们提出一个问题：端子 AA' 和端子 BB' 之间的一对电线是传输线吗？如果是这样，在什么情况下我们应该明确地将这对电线视为传输线，而不是完全忽略它们的存在并将电路视为仅连接到发电机 $\tilde{V}_g$ 的 RC 负载？第一个问题的答案是：是的，

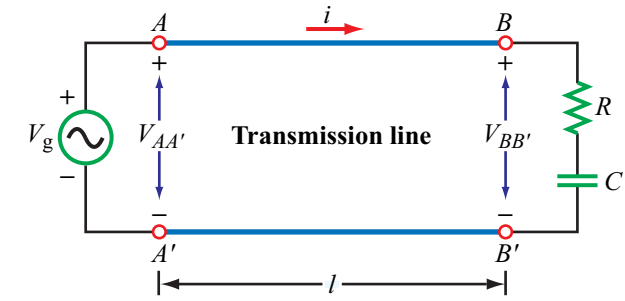


图 2-2 发电机通过长度为 l 的传输线连接到 RC 电路。

the pair of wires does constitute a transmission line. The answer to the second question is: *The factors that determine whether or not we should treat the wires as a transmission line is governed by the length of the line l and the frequency f of the signal provided by the generator.* (As we will see later, the determining factor is the ratio of the length l to the wavelength λ of the wave propagating on the transmission line between the source at terminals AA' and the load at terminals BB' .) If the generator voltage is cosinusoidal in time, then the voltage across the input terminals AA' is

$$V_{AA'} = V_g(t) = V_0 \cos \omega t \quad (\text{V}), \quad (2.1)$$

where $\omega = 2\pi f$ is the angular frequency. If we assume that the current flowing through the wires travels at the speed of light, $c = 3 \times 10^8$ m/s, then the voltage across the output terminals BB' will have to be delayed in time relative to that across AA' by the travel delay time l/c . Thus, assuming no ohmic losses in the transmission line and ignoring other transmission-line effects discussed later in this chapter,

$$V_{BB'}(t) = V_{AA'}(t - l/c) = V_0 \cos[\omega(t - l/c)] = V_0 \cos(\omega t - \phi_0), \quad (2.2)$$

with

$$\phi_0 = \frac{\omega l}{c} \quad (\text{rad}). \quad (2.3)$$

Thus, the time delay associated with the length of the line l manifests itself as a constant phase shift ϕ_0 in the argument of the cosine. Let us compare $V_{BB'}$ with $V_{AA'}$ at $t = 0$ for an ultralow-frequency electronic circuit operating at a frequency $f = 1$ kHz. For a typical wire length $l = 5$ cm, Eqs. (2.1) and (2.2) give $V_{AA'} = V_0$ and $V_{BB'} = V_0 \cos(2\pi f l/c) = 0.99999999998 V_0$. Hence, for all practical purposes, the presence of the transmission line may be ignored, and terminal AA' may be treated as identical with BB' so far as its voltage is concerned. On the other hand, had the line been a 20-km long telephone cable carrying a 1 kHz voice signal, then the same calculation would have led to $V_{BB'} = 0.91 V_0$, which is a deviation of 9%. Furthermore, had the frequency of the signal on the 5-cm long wire been 1.5 GHz instead of 1 kHz, the voltage at the end of the line would have been $V_{BB'} = 0$! Thus, the pair of wires would have a voltage V_0 at AA' and zero at BB' simultaneously. The determining factor is the magnitude of $\phi_0 = \omega l/c$. From Eq. (1.27), the velocity of propagation u_p of a traveling wave is related to the oscillation frequency f and the wavelength λ by

$$u_p = f\lambda \quad (\text{m/s}).$$

In the present case, $u_p = c$. Hence, the phase delay is

$$\phi_0 = \frac{\omega l}{c} = \frac{2\pi f l}{c} = 2\pi \frac{l}{\lambda} \quad \text{radians}. \quad (2.4)$$

► When l/λ is very small, transmission-line effects may be ignored, but when $l/\lambda \gtrsim 0.01$, it may be necessary to account not only for the phase shift due to the time delay but also for the presence of **reflected** signals that may have been bounced back by the load toward the generator. ◀

Power loss on the line and **dispersive** effects may need to be considered as well.

► A dispersive transmission line is one on which the wave velocity is not constant as a function of the frequency f . ◀

This means that the shape of a rectangular pulse, which through Fourier analysis can be decomposed into many sinusoidal waves of different frequencies, gets distorted as it travels down the line because its different frequency components do not propagate at the same velocity (**Fig. 2-3**). Preservation of pulse shape is very important in high-speed data transmission not only between terminals but also across transmission line segments fabricated within high-speed integrated circuits. At 10 GHz, for example, the wavelength is $\lambda = 3$ cm in air but only on the order of 1 cm in a semiconductor material. Hence, even lengths between devices on the order of millimeters become significant, and their presence has to be accounted for in the design of the circuit.

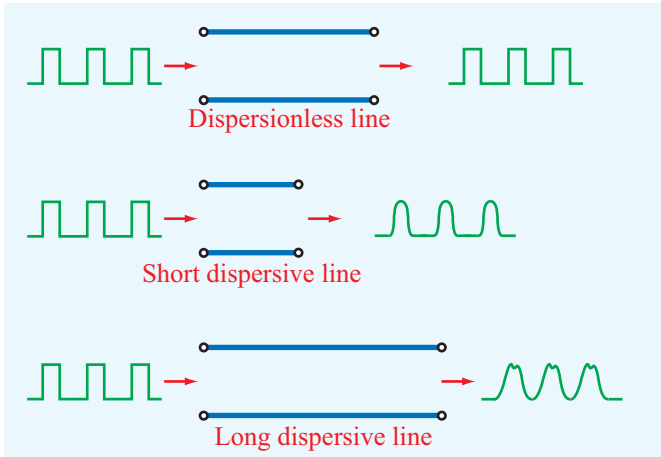


Figure 2-3 A dispersionless line does not distort signals passing through it regardless of its length, whereas a dispersive line distorts the shape of the input pulses because the different frequency components propagate at different velocities. The degree of distortion is proportional to the length of the dispersive line.

这对电线确实构成了传输线。第二个问题的答案是：决定我们是否应该将电线视为传输线的因素取决于线的长度 l 和发生器提供的信号的频率 f 。（正如我们稍后将看到的，决定因素是在端子 AA' 处的源和端子 BB' 处的负载之间的传输线上传播的波的长度 l 与波长 λ 的比率。）如果发电机电压随时间呈余弦曲线，则输入端子 AA' 两端的电压为

$$V_{AA'} = V_g(t) = V_0 \cos \omega t \quad (\text{V}), \quad (2.1)$$

其中 $\omega = 2\pi f$ 是角频率。如果我们假设流过导线的电流以光速 $c = 3 \times 10^8$ m/s 传播，那么输出端子 BB' 两端的电压将必须相对于 AA' 两端的电压在时间上延迟 l/c 。因此，假设传输线中没有欧姆损耗并忽略本章后面讨论的其他传输线效应，

$$V_{BB'}(t) = V_{AA'}(t - l/c) = V_0 \cos[\omega(t - l/c)] = V_0 \cos(\omega t - \phi_0), \quad (2.2)$$

和

$$\phi_0 = \frac{\omega l}{c} \quad (\text{rad}). \quad (2.3)$$

因此，与线 l 的长度相关的时间延迟表现为余弦参数中的恒定相移 ϕ_0 。让我们比较一下 $t = 0$ 时的 $V_{BB'}$ 与 $V_{AA'}$ ，了解以频率 $f = 1$ kHz 运行的超低频电子电路。对于典型的线长 $l = 5$ cm，方程：(2.1) 和 (2.2) 给出 $V_{AA'} = V_0$ 和 $V_{BB'} = V_0 \cos(2\pi f l/c) = 0.99999999998 V_0$ 。因此，出于所有实际目的，可以忽略传输线的存在，并且就其电压而言，端子 AA' 可以被视为与 BB' 相同。另一方面，如果线路是传输 1 kHz 语音信号的 20 公里长的电话电缆，那么相同的计算将导致 $V_{BB'} = 0.91 V_0$ ，偏差为 9%。此外，如果 5 厘米长的电线上的信号频率为 1.5 GHz 而不是 1 kHz，则线路末端的电压将为 $V_{BB'} = 0$ ！因此，这对导线在 AA' 处具有电压 V_0 ，在 BB' 处同时具有零电压。决定因素是 $\phi_0 = \omega l/c$ 的大小。从方程。根据方程 (1.27)，行波的传播速度 u_p 与振荡频率 f 和波长 λ 的关系为：

$$u_p = f\lambda \quad (\text{m/s}).$$

在本例中， $u_p = c$ 。因此，相位延迟为

$$\phi_0 = \frac{\omega l}{c} = \frac{2\pi f l}{c} = 2\pi \frac{l}{\lambda} \quad \text{radians}. \quad (2.4)$$

► 当 l/λ 非常小时，可以忽略传输线效应，但当 $l/\lambda \gtrsim 0.01$ 时，可能不仅需要考虑由于时间延迟而导致的相移，还需要考虑可能已被负载反射回发电机的反射信号的存在。 ◀

线路上的功率损耗和色散效应也可能需要考虑。

► 色散传输线是一种波速作为频率 f 的函数并不恒定的传输线。 ◀

这意味着矩形脉冲的形状（通过傅里叶分析可以分解为许多不同频率的正弦波）在沿线路传播时会发生畸变，因为其不同频率分量不以相同的速度传播（图2-3）。脉冲形状的保持在高速数据传输中非常重要，不仅在终端之间，而且在高速集成电路内制造的传输线段之间也是如此。例如，在 10 GHz 时，空气中的波长为 $\lambda = 3$ cm，但在半导体材料中仅为 1 cm 左右。因此，即使器件之间的长度为毫米级，也变得很重要，并且在电路设计中必须考虑到它们的存在。

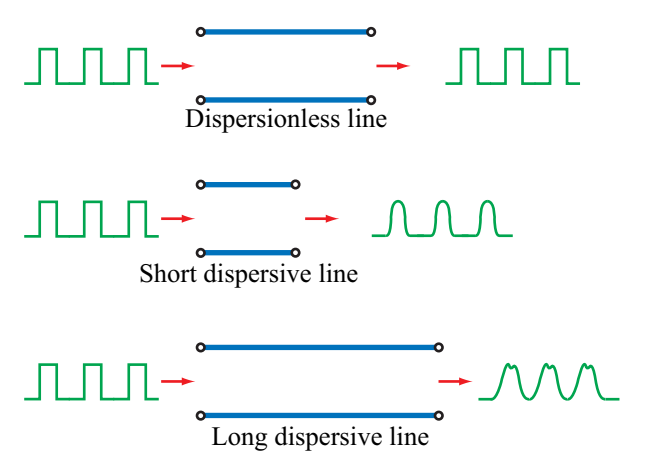


图 2-3 无色散线无论其长度如何都不会扭曲通过它的信号，而色散线会扭曲输入脉冲的形状，因为不同的频率分量以不同的速度传播。畸变程度与色散线的长度成正比。

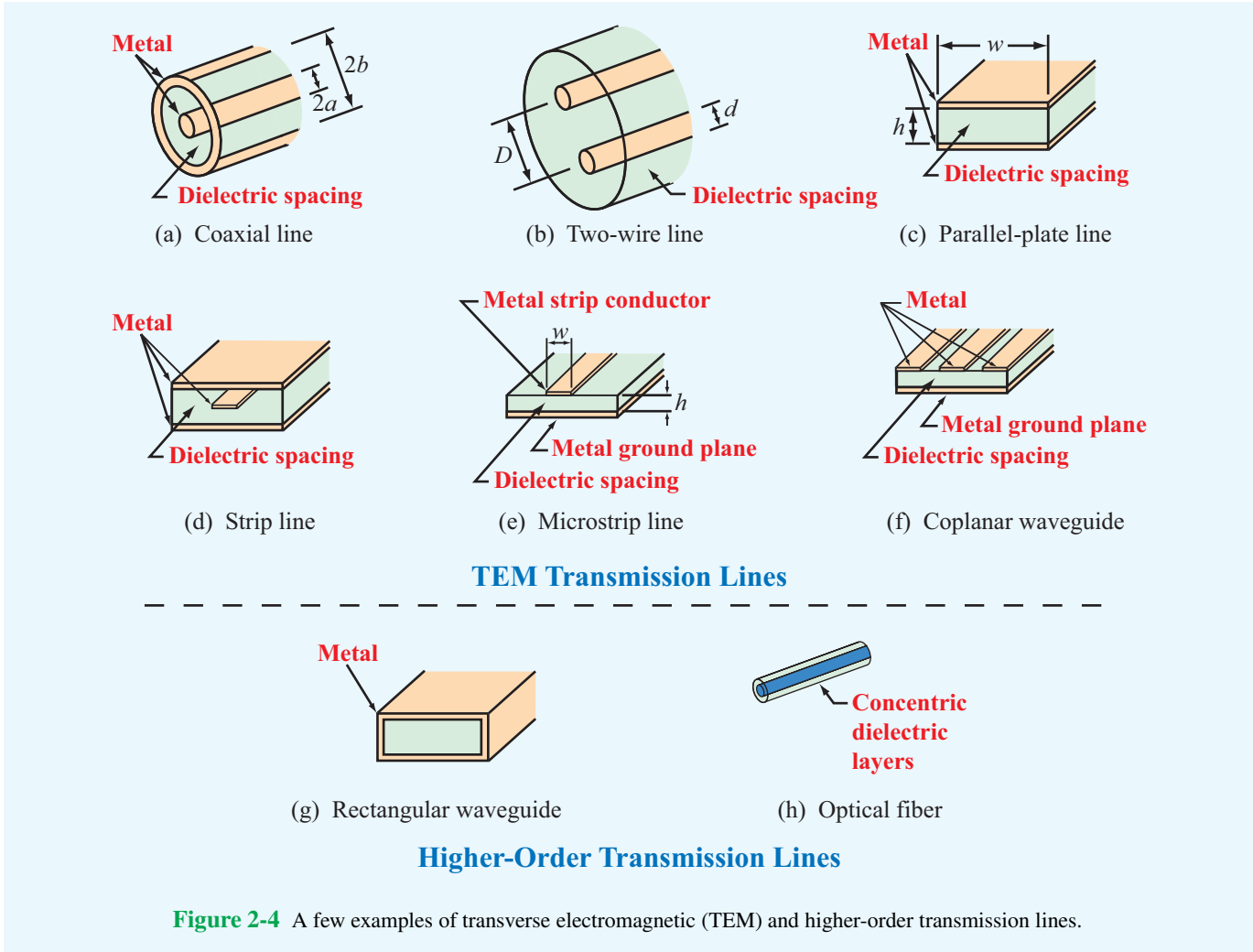


Figure 2-4 A few examples of transverse electromagnetic (TEM) and higher-order transmission lines.

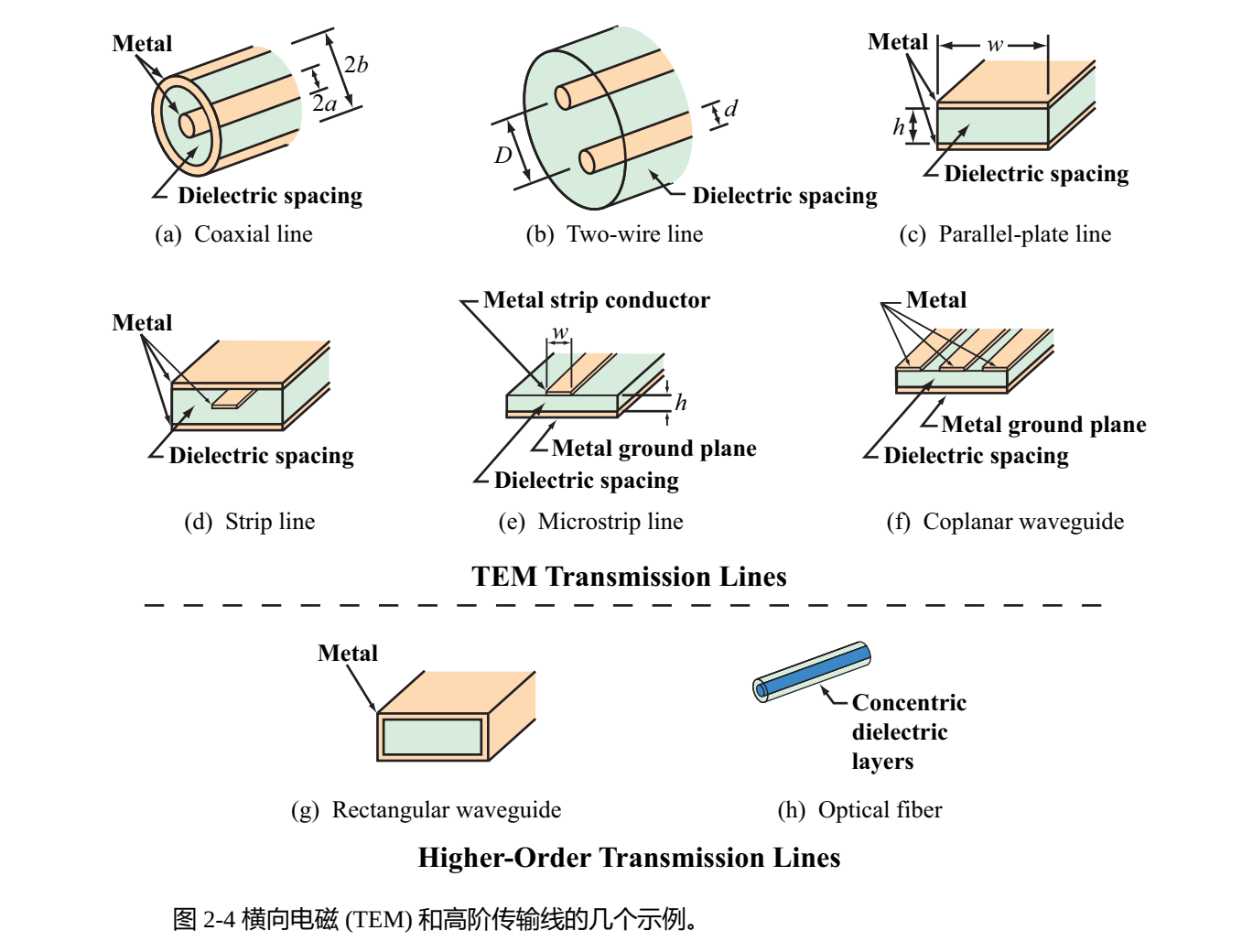


图 2-4 横向电磁 (TEM) 和高阶传输线的几个示例。

2-1.2 Propagation Modes

A few examples of common types of transmission lines are shown in Fig. 2-4. Transmission lines may be classified into two basic types:

- **Transverse electromagnetic (TEM) transmission lines:** Waves propagating along these lines are characterized by electric and magnetic fields that are entirely *transverse* to the direction of propagation. Such an orthogonal configuration is called a **TEM mode**. A good example is the coaxial line shown in Fig. 2-5. The electric field is in the radial direction between the inner and outer conductors, while the magnetic field circles the inner conductor, and neither has a component along the line axis (the direction of wave propagation). Other TEM transmission

lines include the two-wire line and the parallel-plate line (both shown in Fig. 2-4). Although the fields present on a microstrip line do not adhere to the exact definition of a TEM mode, the nontransverse field components are sufficiently small (in comparison with the transverse components) that they may be ignored, thereby allowing the inclusion of microstrip lines in the TEM class. A common feature among TEM lines is that they consist of two parallel conducting surfaces.

- **Higher-order transmission lines:** Waves propagating along these lines have at least one significant field component in the direction of propagation. Hollow conducting waveguides, dielectric rods, and optical fibers belong to this class of lines (Chapter 8).

2-1.2 传播模式

图 2-4 显示了一些常见类型的传输线示例。传输线可分为两种基本类型：

- **横向电磁 (TEM) 传输线：**沿这些线传播的波的特点是电场和磁场完全横向于传播方向。这种正交配置称为 TEM 模式。图 2-5 所示的同轴线就是一个很好的例子。电场沿内导体和外导体之间的径向方向，而磁场围绕内导体，并且两者都没有沿线轴（波传播方向）的分量。其他 TEM 传输

线路包括双线线路和平行板线路（均如图 2-4 所示）。尽管微带线上的场不符合 TEM 模式的精确定义，但非横向场分量足够小（与横向分量相比），可以忽略它们，从而允许将微带线包含在 TEM 类中。TEM 线的一个共同特征是它们由两个平行的导电表面组成。

- **高阶传输线：**沿这些线传播的波在传播方向上至少有一个重要的场分量。空心传导波导、介电棒和光纤都属于此类线路（第 8 章）。

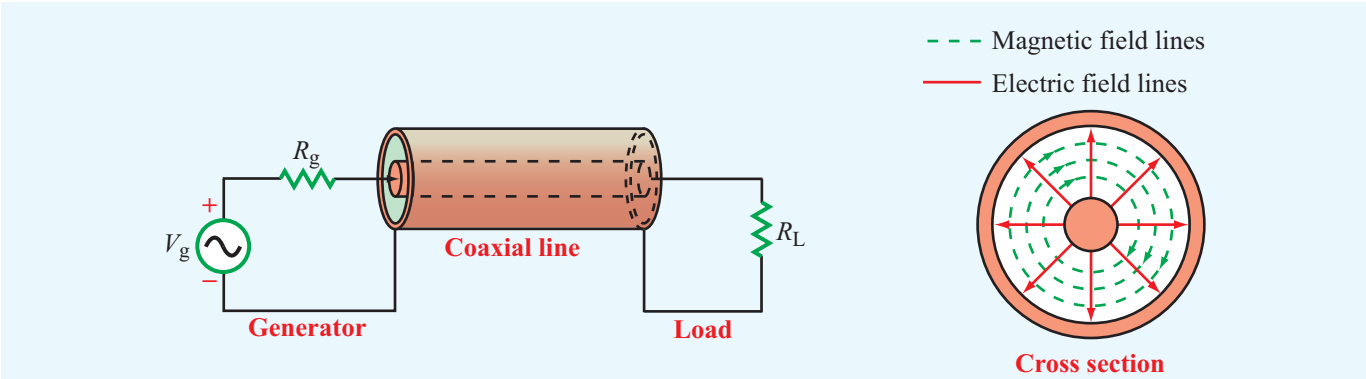


Figure 2-5 In a coaxial line, the electric field is in the radial direction between the inner and outer conductors, and the magnetic field forms circles around the inner conductor. The coaxial line is a transverse electromagnetic (TEM) transmission line because both the electric and magnetic fields are orthogonal to the direction of propagation between the generator and the load.

Only TEM-mode transmission lines are treated in this chapter. This is because they are more commonly used in practice, and fortunately, less mathematical rigor is required for treating them than is required for lines that support higher-order modes. We start our treatment by representing the transmission line in terms of a lumped-element circuit model, and then we apply Kirchhoff's voltage and current laws to derive a pair of equations governing their behavior, known as the *telegrapher's equations*. By combining these equations, we obtain wave equations for the voltage and current at any location along the line. Solution of the wave equations for the sinusoidal steady-state case leads to a set of formulas that can be used to solve a wide range of practical problems. In the latter part of this chapter, we introduce a graphical tool known as the *Smith chart*, which facilitates the solution of transmission-line problems without having to perform laborious calculations involving complex numbers.

2-2 Lumped-Element Model

When we draw a schematic of an electronic circuit, we use specific symbols to represent resistors, capacitors, inductors, diodes, and the like. In each case, the symbol represents the functionality of the device rather than its shape, size, or other attributes. We shall do the same for transmission lines.

► A transmission line will be represented by a parallel-wire configuration (Fig. 2-6(a)), regardless of its specific shape or constitutive parameters. ◀

Thus, Fig. 2-6(a) may represent a coaxial line, a two-wire line, or any other TEM line. Drawing again on our familiarity with electronic circuits, when we analyze a circuit containing a transistor, we mimic the functionality of the transistor by an equivalent circuit composed of sources, resistors, and capacitors. We apply the same approach to the transmission line by orienting the line along the z direction, subdividing it into differential sections each of length Δz (Fig. 2-6(b)), and then representing each section by an equivalent circuit, as illustrated in Fig. 2-6(c). This representation, often called the *lumped-element circuit* model, consists of four basic elements with values that henceforth will be called the *transmission line parameters*. These are:

- R' : The combined *resistance* of both conductors per unit length, in Ω/m ,
- L' : The combined *inductance* of both conductors per unit length, in H/m ,
- G' : The *conductance* of the insulation medium between the two conductors per unit length, in S/m , and
- C' : The *capacitance* of the two conductors per unit length, in F/m .

Whereas the four line parameters are characterized by different formulas for different types of transmission lines, the equivalent model represented by Fig. 2-6(c) is equally applicable to all TEM transmission lines. *The prime superscript is used as a reminder that the line parameters are differential quantities whose units are per unit length.* Expressions for the line parameters R' , L' , G' , and C' are given in Table 2-1 for the three types of TEM transmission lines diagrammed in parts (a) through (c) of Fig. 2-4. For

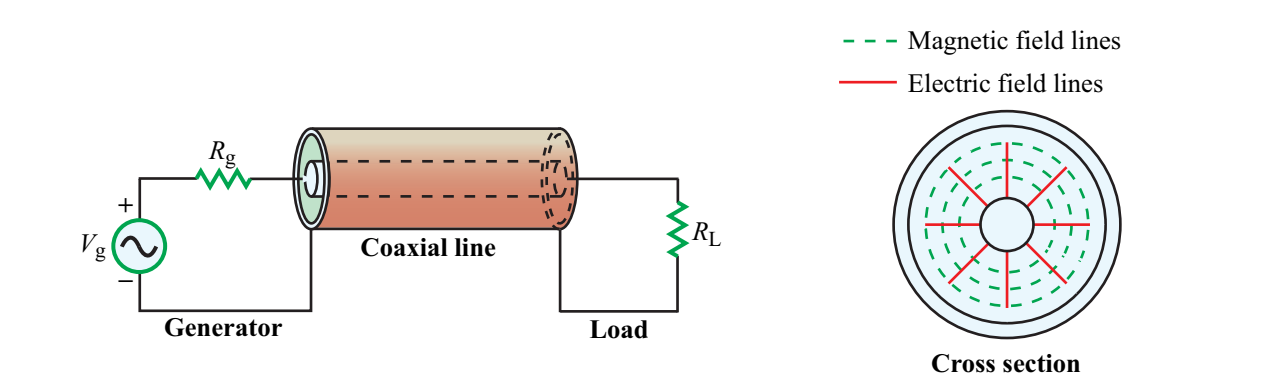


图2-5 在同轴线中，电场沿内导体和外导体之间的径向方向，磁场围绕内导体形成圆圈。同轴线是横向电磁 (TEM) 传输线，因为电场和磁场都与发电机和负载之间的传播方向正交。

本章仅讨论 TEM 模式传输线。这是因为它们在实践中更常用，幸运的是，处理它们所需的数学严谨性比支持高阶模式的线所需的数学严谨性要少。我们首先用集总元件电路模型表示传输线，然后应用基尔霍夫电压和电流定律推导出对控制其行为的方程，称为电报员方程。通过组合这些方程，我们可以获得沿线任意位置的电压和电流的波动方程。正弦稳态情况下波动方程的求解得出了一组可用于解决各种实际问题的公式。在本章的后半部分，我们将介绍一种称为史密斯圆图的图形工具，它有助于解决传输线问题，而无需执行涉及复数的繁琐计算。

2-2 集总元件模型

当我们绘制电子电路原理图时，我们使用特定的符号来表示电阻器、电容器、电感器、二极管等。在每种情况下，符号都代表设备的功能，而不是其形状、大小或其他属性。我们将对传输线做同样的事情。

► 传输线将由平行线配置表示 (图2-6 (a))，无论其具体形状或本构参数如何。 ◀

因此，图2-6(a)可以表示同轴线、两线或任何其他TEM线。再次利用我们对电子电路的熟悉，当我们分析包含晶体管的电路时，我们通过由源、电阻器和电容器组成的等效电路来模拟晶体管的功能。我们将相同的方法应用于传输线，将传输线沿 z 方向定向，将其细分为每个长度为 Δz 的差分部分 (图 2-6 (b))，然后用等效电路表示每个部分，如图 2-6 (c) 所示。这种表示法通常称为集总元件电路模型，由四个基本元件组成，其值今后将称为传输线参数。这些都是：

- R' : 单位长度两个导体的组合电阻，单位为 Ω/m ,
- L' : 每单位长度两个导体的组合电感，单位为 H/m ,
- G' : 单位长度两导体之间绝缘介质的电导率，单位为 S/m ,
- C' : 单位长度上两个导体的电容，单位为 F/m 。

尽管对于不同类型的传输线，四个线参数有不同的公式来表征，但图 2-6 (c) 所示的等效模型同样适用于所有 TEM 传输线。主上标用于提醒，线参数是微分量，其单位为每单位长度。

对于图2-4(a)至(c)中所示的三种类型的TEM传输线，表 2-1给出了线参数 R' 、 L' 、 G' 和 C' 的表达式。为了

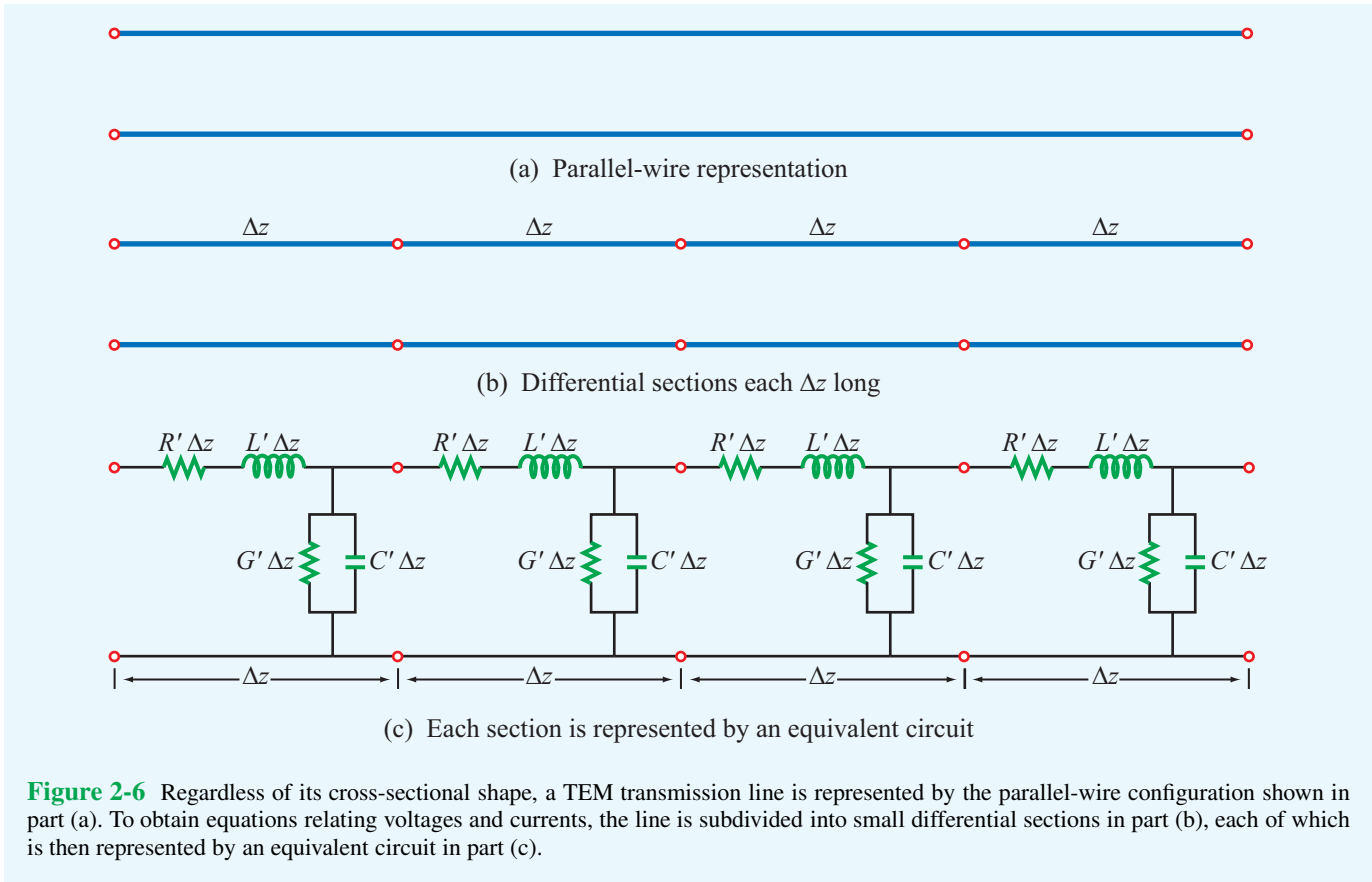


Figure 2-6 Regardless of its cross-sectional shape, a TEM transmission line is represented by the parallel-wire configuration shown in part (a). To obtain equations relating voltages and currents, the line is subdivided into small differential sections in part (b), each of which is then represented by an equivalent circuit in part (c).

each of these lines, the expressions are functions of two sets of parameters: (1) geometric parameters defining the cross-sectional dimensions of the given line and (2) the electromagnetic constitutive parameters of the conducting and insulating materials. The pertinent geometric parameters are:

• **Coaxial line (Fig. 2-4(a)):**

a = outer radius of inner conductor, m
 b = inner radius of outer conductor, m

• **Two-wire line (Fig. 2-4(b)):**

d = diameter of each wire, m
 D = spacing between wires' centers, m

• **Parallel-plate line (Fig. 2-4(c)):**

w = width of each plate, m
 h = thickness of insulation between plates, m

► The pertinent **constitutive parameters** apply to all three lines and consist of two groups:

- (1) μ_c and σ_c are the magnetic permeability and electrical conductivity of the conductors, and
- (2) ϵ , μ , and σ are the electrical permittivity, magnetic permeability, and electrical conductivity of the insulation material separating them. ◀

Appendix B contains tabulated values for these constitutive parameters for various materials. For the purposes of this chapter, we need not concern ourselves with the derivations leading to the expressions in **Table 2-1**. The techniques necessary for computing R' , L' , G' , and C' for the general case of an arbitrary two-conductor configuration are presented in later chapters.

The lumped-element model shown in **Fig. 2-6(c)** reflects the physical phenomena associated with the currents and voltages on any TEM transmission line. It consists of two in-series elements, R' and L' , and two shunt elements, G' and C' . To

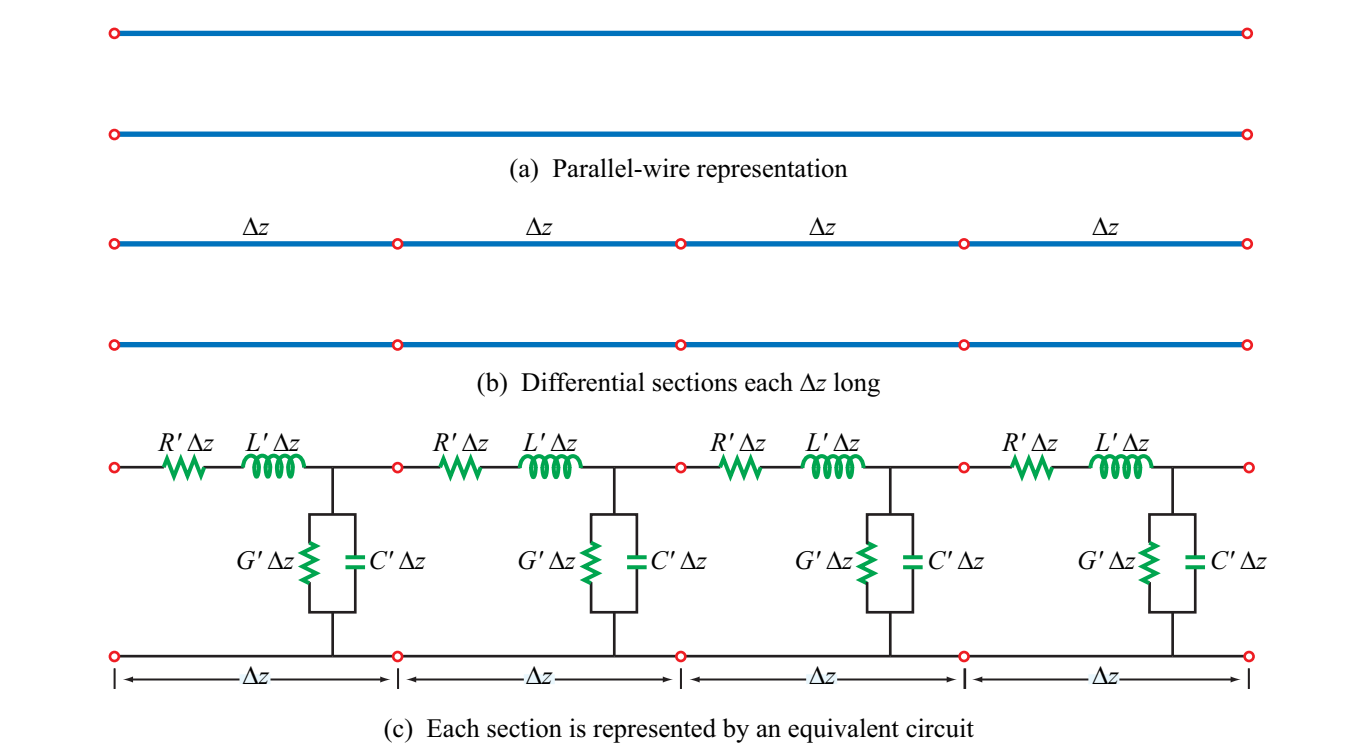


图 2-6 无论其横截面形状如何，TEM 传输线都由 (a) 部分所示的平行线配置表示。为了获得与电压和电流相关的方程，在 (b) 部分中将线路细分为小的微分部分，然后每个部分由 (c) 部分中的等效电路表示。

对于每条线，表达式都是两组参数的函数：(1) 定义给定线的横截面尺寸的几何参数和 (2) 导电和绝缘材料的电磁本构参数。相关的几何参数是：

• 同轴线 (图2-4 (a)) :

a = 内导体外半径, m b = 外导体内半径, m

• 两线制线路 (图2-4 (b)) :

d = 每根导线的直径, m D = 导线中心之间的间距, m

• 平行板线 (图2-4 (c)) :

w = 每块板的宽度, m h = 板之间绝缘层的厚度, m

► 相关的本构参数适用于所有三条线，并由两组组成：

(1) μ_c 和 σ_c 是导体的磁导率和电导率，并且

(2) ϵ 、 μ 和 σ 是分隔它们的绝缘材料的介电常数、磁导率和电导率。 ◀

附录 B 包含各种材料的这些本构参数的表格值。为了本章的目的，我们不需要关心表 2-1 中表达式的推导。对于任意两导体配置的一般情况，计算 R' 、 L' 、 G' 和 C' 所需的技术将在后面的章节中介绍。

图 2-6(c) 所示的集总元件模型反映了与任何 TEM 传输线上的电流和电压相关的物理现象。它由两个串联元件 R' 和 L' 以及两个并联元件 G' 和 C' 组成。到

Table 2-1 Transmission-line parameters R' , L' , G' , and C' for three types of lines.

Parameter	Coaxial	Two-Wire	Parallel-Plate	Unit
R'	$\frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L'	$\frac{\mu}{2\pi} \ln(b/a)$	$\frac{\mu}{\pi} \ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]$	$\frac{\mu h}{w}$	H/m
G'	$\frac{2\pi\sigma}{\ln(b/a)}$	$\frac{\pi\sigma}{\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]}$	$\frac{\sigma w}{h}$	S/m
C'	$\frac{2\pi\epsilon}{\ln(b/a)}$	$\frac{\pi\epsilon}{\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]}$	$\frac{\epsilon w}{h}$	F/m

Notes: (1) Refer to Fig. 2-4 for definitions of dimensions. (2) μ , ϵ , and σ pertain to the insulating material between the conductors. (3) $R_s = \sqrt{\pi f \mu_c / \sigma_c}$. (4) μ_c and σ_c pertain to the conductors. (5) If $(D/d)^2 \gg 1$, then $\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right] \approx \ln(2D/d)$.

explain the lumped-element model, consider a small section of a coaxial line, as shown in Fig. 2-7. The line consists of inner and outer conductors of radii a and b separated by a material with permittivity ϵ , permeability μ , and conductivity σ . The two metal conductors are made of a material with conductivity σ_c and permeability μ_c .

Resistance R'

When a voltage source is connected across the terminals connected to the two conductors at the sending end of the line, currents flow through the conductors, primarily along the outer surface of the inner conductor and the inner surface of the outer conductor. The line resistance R' accounts for the combined resistance per unit length of the inner and outer conductors. The expression for R' is derived in Chapter 7 and is given by Eq. (7.96) as

$$R' = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\text{coax line}) \quad (\Omega/\text{m}), \quad (2.5)$$

where R_s , which represents the **surface resistance** of the conductors, is given by Eq. (7.92a) as

$$R_s = \sqrt{\frac{\pi f \mu_c}{\sigma_c}} \quad (\Omega). \quad (2.6)$$

The surface resistance depends not only on the material properties of the conductors (σ_c and μ_c), but also on the frequency f of the wave traveling on the line.

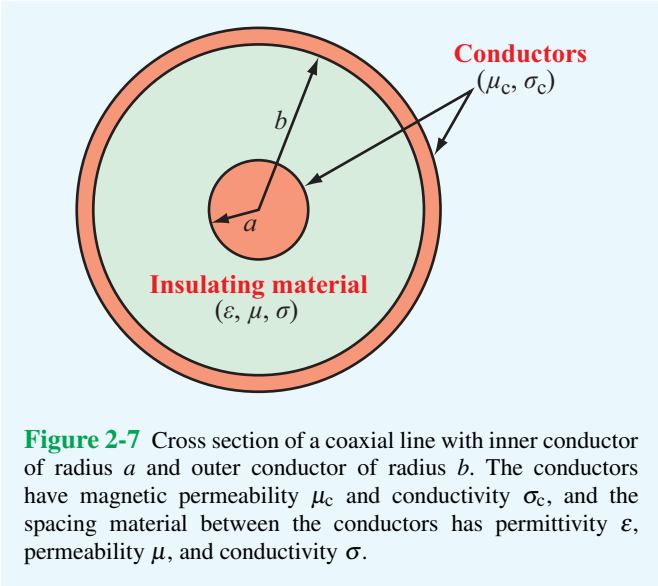


Figure 2-7 Cross section of a coaxial line with inner conductor of radius a and outer conductor of radius b . The conductors have magnetic permeability μ_c and conductivity σ_c , and the spacing material between the conductors has permittivity ϵ , permeability μ , and conductivity σ .

► For a **perfect conductor** with $\sigma_c = \infty$ or a high-conductivity material such that $(f \mu_c / \sigma_c) \ll 1$, R_s approaches zero, and so does R' . ◀

表 2-1 三种类型线路的传输线参数 R' 、 L' 、 G' 和 C' 。

Parameter	Coaxial	Two-Wire	Parallel-Plate	Unit
R'	$\frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$	$\frac{2R_s}{\pi d}$	$\frac{2R_s}{w}$	Ω/m
L'	$\frac{\mu}{2\pi} \ln(b/a)$	$\frac{\mu}{\pi} \ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]$	$\frac{\mu h}{w}$	H/m
G'	$\frac{2\pi\sigma}{\ln(b/a)}$	$\frac{\pi\sigma}{\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]}$	$\frac{\sigma w}{h}$	S/m
C'	$\frac{2\pi\epsilon}{\ln(b/a)}$	$\frac{\pi\epsilon}{\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right]}$	$\frac{\epsilon w}{h}$	F/m

Notes: (1) Refer to Fig. 2-4 for definitions of dimensions. (2) μ , ϵ , and σ pertain to the insulating material between the conductors. (3) $R_s = \sqrt{\pi f \mu_c / \sigma_c}$. (4) μ_c and σ_c pertain to the conductors. (5) If $(D/d)^2 \gg 1$, then $\ln \left[(D/d) + \sqrt{(D/d)^2 - 1} \right] \approx \ln(2D/d)$.

为了解释集总元件模型，考虑同轴线的一小部分，如图 2-7 所示。该线路由半径为 a 和 b 的内导体和外导体组成，由具有介电常数 ϵ 、磁导率 μ 和电导率 σ 的材料分隔开。两个金属导体由电导率 σ_c 和磁导率 μ_c 的材料制成。

电阻 R'

当电压源跨接在线路发送端连接到两个导体的端子上时，电流主要沿着内导体的外表面和外导体的内表面流过导体。线路电阻 R' 是内导体和外导体每单位长度的组合电阻。 R' 的表达式在第 7 章中推导，由式 (1) 给出。(7.96) 作为

$$R' = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\text{coax line}) \quad (\Omega/\text{m}), \quad (2.5)$$

其中 R_s 表示导体的表面电阻，由方程 1 给出。(7.92a) 为

$$R_s = \sqrt{\frac{\pi f \mu_c}{\sigma_c}} \quad (\Omega). \quad (2.6)$$

表面电阻不仅取决于导体的材料特性 (σ_c 和 μ_c)，还取决于在线路上传播的波的频率 f 。

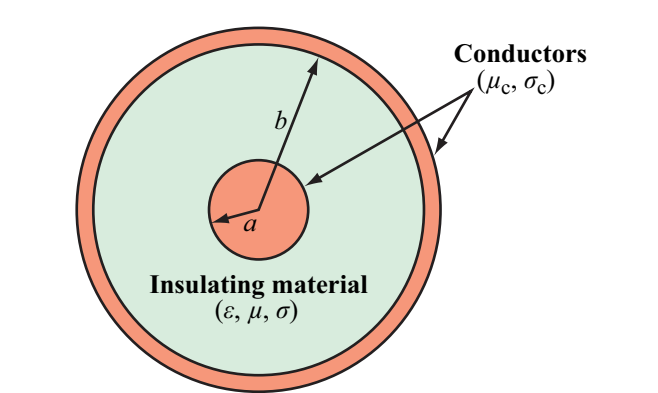


图 2-7 内导体半径为 a 、外导体半径为 b 的同轴线横截面。导体具有磁导率 μ_c 和电导率 σ_c ，并且导体之间的间隔材料具有介电常数 ϵ 、磁导率 μ 和电导率 σ 。

► 对于具有 $\sigma_c = \infty$ 的理想导体或 $(f \mu_c / \sigma_c) \ll 1$ 的高电导率材料， R_s 接近零， R' 也接近零。 ◀

Inductance L'

Next, let us examine the line inductance L' , which accounts for the joint inductance of both conductors. Application of Ampère’s law in Chapter 5 to the definition of inductance leads to the following expression [Eq. (5.99)] for the inductance per unit length of a coaxial line:

$$L' = \frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right) \quad \textbf{(coax line)} \quad (\text{H/m}). \quad (2.7)$$

Conductance G'

The line conductance G' accounts for current flow between the outer and inner conductors, made possible by the conductivity σ of the insulator. It is precisely because the current flow is from one conductor to the other that G' appears as a *shunt* element in the lumped-element model. For the coaxial line, the conductance per unit length is given by Eq. (4.86) as

$$G' = \frac{2\pi\sigma}{\ln(b/a)} \quad \textbf{(coax line)} \quad (\text{S/m}). \quad (2.8)$$

► If the material separating the inner and outer conductors is a *perfect dielectric* with $\sigma = 0$, then $G' = 0$. ◀

Capacitance C'

The last line parameter on our list is the line capacitance C' . When equal and opposite charges are placed on any two non-contacting conductors, a voltage difference develops between them. Capacitance is defined as the ratio of the charge to the voltage difference. For the coaxial line, the capacitance per unit length is given by Eq. (4.127) as

$$C' = \frac{2\pi\epsilon}{\ln(b/a)} \quad \textbf{(coax line)} \quad (\text{F/m}). \quad (2.9)$$

All TEM transmission lines share the following useful relations:

$$L'C' = \mu\epsilon \quad \textbf{(all TEM lines)}, \quad (2.10)$$

and

$$\frac{G'}{C'} = \frac{\sigma}{\epsilon} \quad \textbf{(all TEM lines)}. \quad (2.11)$$

If the insulating medium between the conductors is air, the transmission line is called an *air line* (e.g., coaxial air line or

two-wire air line). For an air line, $\epsilon = \epsilon_0 = 8.854 \times 10^{-12}$ F/m, $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m, $\sigma = 0$, and $G' = 0$.

Concept Question 2-1: What is a transmission line? When should transmission-line effects be considered, and when may they be ignored?

Concept Question 2-2: What is the difference between dispersive and nondispersive transmission lines? What is the practical significance of dispersion?

Concept Question 2-3: What constitutes a TEM transmission line?

Concept Question 2-4: What purpose does the lumped-element circuit model serve? How are the line parameters R' , L' , G' , and C' related to the physical and electromagnetic constitutive properties of the transmission line?

Exercise 2-1: Use **Table 2-1** to evaluate the line parameters of a two-wire air line with wires of radius 1 mm, separated by a distance of 2 cm. The wires may be treated as perfect conductors with $\sigma_c = \infty$.

Answer: $R' = 0$, $L' = 1.20$ ($\mu\text{H/m}$), $G' = 0$, $C' = 9.29$ (pF/m). (See [Ex. 2-1](#).)

Exercise 2-2: Calculate the transmission line parameters at 1 MHz for a coaxial air line with inner and outer conductor diameters of 0.6 cm and 1.2 cm, respectively. The conductors are made of copper (see Appendix B for μ_c and σ_c of copper).

Answer: $R' = 2.07 \times 10^{-2}$ (Ω/m), $L' = 0.14$ ($\mu\text{H/m}$), $G' = 0$, $C' = 80.3$ (pF/m). (See [Ex. 2-2](#).)

2-3 Transmission-Line Equations

A transmission line usually connects a source on one end to a load on the other. Before considering the complete circuit, however, we will develop general equations that describe the voltage across and current carried by the transmission line as a function of time t and spatial position z . Using the lumped-element model of **Fig. 2-6(c)**, we begin by considering a differential length Δz as shown in **Fig. 2-8**. The quantities $v(z,t)$ and $i(z,t)$ denote the instantaneous voltage and current at the left end of the differential section (node N), and similarly $v(z + \Delta z,t)$ and $i(z + \Delta z,t)$ denote the same quantities at node

电感 L'

接下来，让我们检查线路电感 L' ，它是两个导体的联合电感。应用第 5 章中的安培环路定律来定义电感，可得出以下表达式 [Eq.1]。 (5.99)] 同轴线单位长度的电感：

$$L' = \frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right) \quad \textbf{(coax line)} \quad (\text{H/m}). \quad (2.7)$$

电导 G'

线路电导 G' 说明了外导体和内导体之间的电流，这是由绝缘体的电导率 σ 实现的。正是因为电流从一个导体流向另一个导体， G' 在集总元件模型中显示为分流元件。对于同轴线，单位长度的电导由式 (1) 给出。 (4.86) 作为

$$G' = \frac{2\pi\sigma}{\ln(b/a)} \quad \textbf{(coax line)} \quad (\text{S/m}). \quad (2.8)$$

► 如果分隔内导体和外导体的材料是理想电介质且 $\sigma = 0$ ，则 $G' = 0$ 。 ◀

电容 C'

我们列表中的最后一个线路参数是线路电容 C' 。当相等且相反的电荷放置在任何两个非接触导体上时，它们之间就会产生电压差。电容定义为电荷与电压差之比。对于同轴线，单位长度的电容由式 (1) 给出。 (4.127) 作为

$$C' = \frac{2\pi\epsilon}{\ln(b/a)} \quad \textbf{(coax line)} \quad (\text{F/m}). \quad (2.9)$$

所有 TEM 传输线都具有以下有用关系：

$$L'C' = \mu\epsilon \quad \textbf{(all TEM lines)}, \quad (2.10)$$

和

$$\frac{G'}{C'} = \frac{\sigma}{\epsilon} \quad \textbf{(all TEM lines)}. \quad (2.11)$$

如果导体之间的绝缘介质是空气，则传输线称为空气线（例如同轴空气线或

两线空气管路）。对于航空公司， $\epsilon = \epsilon_0 = 8.854 \times 10^{-12}$ F/m, $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m, $\sigma = 0$ 和 $G' = 0$ 。

概念问题2-1：什么是传输线？什么时候应该考虑传输线效应，什么时候可以忽略它们？

概念问题 2-2：色散传输线和非色散传输线有什么区别？色散的实际意义是什么？

概念问题 2-3：TEM 传输线由什么构成？

概念问题 2-4：集总元件电路模型有什么用途？线路参数 R' 、 L' 、 G' 和 C' 与传输线的物理和电磁本构特性有何关系？

练习 2-1：使用表 2-1 评估两线空气管路的线路参数，该两线空气管路的导线半径为 1 毫米，间隔距离为 2 厘米。电线可以被视为具有 $\sigma_c = \infty$ 的理想导体。

答案： $R' = 0$ 、 $L' = 1.20$ ($\mu\text{H/m}$)、 $G' = 0$ 、 $C' = 9.29$ (pF/m)。 （参见 [Ex. 2-1](#)。）

练习 2-2：计算内导体直径和外导体直径分别为 0.6 厘米和 1.2 厘米的同轴空气线路在 1 MHz 时的传输线参数。导体由铜制成（有关铜的 μ_c 和 σ_c ，请参阅附录 B）。

答案： $R' = 2.07 \times 10^{-2}$ (Ω/m)， $L' = 0.14$ ($\mu\text{H/m}$)， $G' = 0$ ， $C' = 80.3$ (pF/m)。 （参见 [Ex. 2-2](#)。）

2-3 传输线方程

传输线通常将一端的源连接到另一端的负载。然而，在考虑完整电路之前，我们将建立一般方程，将传输线上的电压和传输线承载的电流描述为时间 t 和空间位置 z 的函数。使用图 2-6(c) 的集总元件模型，我们首先考虑如图 2-8 所示的微分长度 Δz 。量 $v(z,t)$ 和 $i(z,t)$ 表示差分部分左端（节点 N ）处的瞬时电压和电流，类似地， $v(z + \Delta z,t)$ 和 $i(z + \Delta z,t)$ 表示节点处的相同量

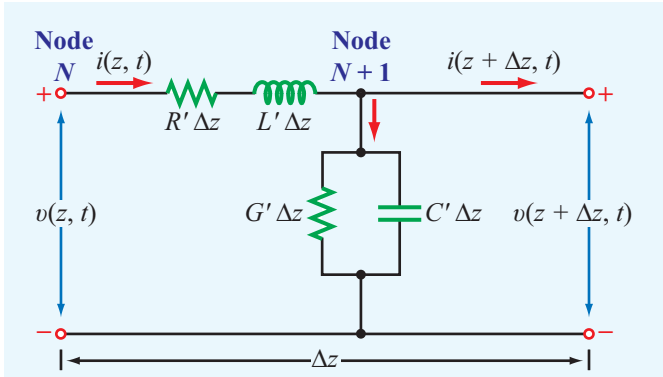


Figure 2-8 Equivalent circuit of a two-conductor transmission line of differential length Δz .

Except for the last section of this chapter, our primary interest is in sinusoidal steady-state conditions. To that end, we make use of the phasor representation with a cosine reference, as outlined in Section 1-7. Thus, we define

$$v(z, t) = \Re[\tilde{V}(z) e^{j\omega t}], \quad (2.17a)$$

$$i(z, t) = \Re[\tilde{I}(z) e^{j\omega t}], \quad (2.17b)$$

where $\tilde{V}(z)$ and $\tilde{I}(z)$ are the phasor counterparts of $v(z, t)$ and $i(z, t)$, respectively, each of which may be real or complex. Upon substituting Eqs. (2.17a) and (2.17b) into Eqs. (2.14) and (2.16), and utilizing the property given by Eq. (1.62) that $\partial/\partial t$ in the time domain is equivalent to multiplication by $j\omega$ in the phasor domain, we obtain the following pair of equations:

$$-\frac{d\tilde{V}(z)}{dz} = (R' + j\omega L') \tilde{I}(z), \quad (2.18a)$$

$$-\frac{d\tilde{I}(z)}{dz} = (G' + j\omega C') \tilde{V}(z). \quad (2.18b)$$

(telegrapher's equations in phasor form)

2-4 Wave Propagation on a Transmission Line

The two first-order coupled equations (2.18a) and (2.18b) can be combined to give two second-order uncoupled wave equations: one for $\tilde{V}(z)$ and another for $\tilde{I}(z)$. The wave equation for $\tilde{V}(z)$ is derived by first differentiating both sides of Eq. (2.18a) with respect to z , resulting in

$$-\frac{d^2\tilde{V}(z)}{dz^2} = (R' + j\omega L') \frac{d\tilde{I}(z)}{dz}. \quad (2.19)$$

Then, upon substituting Eq. (2.18b) for $d\tilde{I}(z)/dz$, Eq. (2.19) becomes

$$\frac{d^2\tilde{V}(z)}{dz^2} - (R' + j\omega L')(G' + j\omega C') \tilde{V}(z) = 0, \quad (2.20)$$

or

$$\frac{d^2\tilde{V}(z)}{dz^2} - \gamma^2 \tilde{V}(z) = 0, \quad (2.21)$$

(wave equation for $\tilde{V}(z)$)

where

$$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}. \quad (2.22)$$

(propagation constant)

($N + 1$), located at the right end of the section. Application of Kirchhoff's voltage law accounts for the voltage drop across the series resistance $R'\Delta z$ and inductance $L'\Delta z$:

$$v(z, t) - R'\Delta z i(z, t) - L'\Delta z \frac{\partial i(z, t)}{\partial t} - v(z + \Delta z, t) = 0. \quad (2.12)$$

Upon dividing all terms by Δz and rearranging them, we obtain

$$-\left[\frac{v(z + \Delta z, t) - v(z, t)}{\Delta z}\right] = R' i(z, t) + L' \frac{\partial i(z, t)}{\partial t}. \quad (2.13)$$

In the limit as $\Delta z \rightarrow 0$, Eq. (2.13) becomes a differential equation:

$$-\frac{\partial v(z, t)}{\partial z} = R' i(z, t) + L' \frac{\partial i(z, t)}{\partial t}. \quad (2.14)$$

Similarly, Kirchhoff's current law accounts for current drawn from the upper line at node ($N + 1$) by the parallel conductance $G'\Delta z$ and capacitance $C'\Delta z$:

$$i(z, t) - G'\Delta z v(z + \Delta z, t) - C'\Delta z \frac{\partial v(z + \Delta z, t)}{\partial t} - i(z + \Delta z, t) = 0. \quad (2.15)$$

Upon dividing all terms by Δz and taking the limit $\Delta z \rightarrow 0$, Eq. (2.15) becomes a second-order differential equation:

$$-\frac{\partial i(z, t)}{\partial z} = G' v(z, t) + C' \frac{\partial v(z, t)}{\partial t}. \quad (2.16)$$

The first-order differential equations (2.14) and (2.16) are the time-domain forms of the **transmission-line equations**, known as the **telegrapher's equations**.

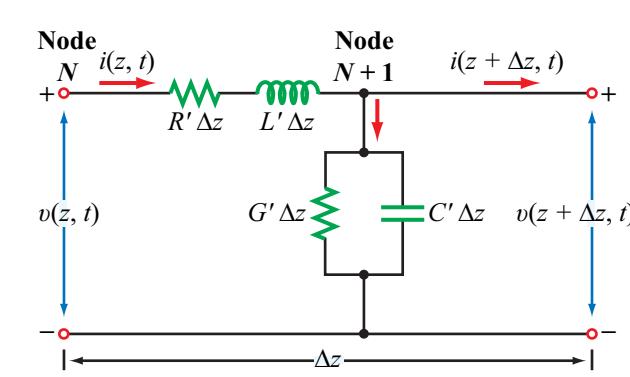


图 2-8 差分长度 Δz 的两导体传输线的等效电路。

($N + 1$), 位于该部分的右端。应用基尔霍夫电压定律可计算串联电阻 $R'\Delta z$ 和电感 $L'\Delta z$ 两端的压降:

$$v(z, t) - R'\Delta z i(z, t) - L'\Delta z \frac{\partial i(z, t)}{\partial t} - v(z + \Delta z, t) = 0. \quad (2.12)$$

将所有项除以 Δz 并重新排列它们后, 我们得到

$$-\left[\frac{v(z + \Delta z, t) - v(z, t)}{\Delta z}\right] = R' i(z, t) + L' \frac{\partial i(z, t)}{\partial t}. \quad (2.13)$$

在极限为 $\Delta z \rightarrow 0$ 时, 式: (2.13) 变为微分方程:

$$-\frac{\partial v(z, t)}{\partial z} = R' i(z, t) + L' \frac{\partial i(z, t)}{\partial t}. \quad (2.14)$$

类似地, 基尔霍夫电流定律解释了通过并联电导 $G'\Delta z$ 和电容 $C'\Delta z$ 从节点 ($N + 1$) 处的上部线汲取的电流:

$$i(z, t) - G'\Delta z v(z + \Delta z, t) - C'\Delta z \frac{\partial v(z + \Delta z, t)}{\partial t} - i(z + \Delta z, t) = 0. \quad (2.15)$$

将所有项除以 Δz 并取极限 $\Delta z \rightarrow 0$ 时, 方程: (2.15) 变为二阶微分方程:

$$-\frac{\partial i(z, t)}{\partial z} = G' v(z, t) + C' \frac{\partial v(z, t)}{\partial t}. \quad (2.16)$$

一阶微分方程 (2.14) 和 (2.16) 是传输线方程的时域形式, 称为电报员方程。

除了本章的最后一节之外, 我们的主要兴趣是正弦稳态条件。为此, 我们使用带有余弦参考的相量表示, 如第 1-7 节所述。因此, 我们定义

$$v(z, t) = \Re[\tilde{V}(z) e^{j\omega t}], \quad (2.17a)$$

$$i(z, t) = \Re[\tilde{I}(z) e^{j\omega t}], \quad (2.17b)$$

其中, $\tilde{V}(z)$ 和 $\tilde{I}(z)$ 分别是 $v(z, t)$ 和 $i(z, t)$ 的相量对应项, 其中每个都可以是实数或复数。代入等式后。 (2.17a) 和 (2.17b) 代入等式。 (2.14) 和 (2.16), 并利用等式给出的属性。 (1.62) 时域中的 $\partial/\partial t$ 等价于相量域中乘以 $j\omega$, 我们得到如下一对方程:

$$-\frac{d\tilde{V}(z)}{dz} = (R' + j\omega L') \tilde{I}(z), \quad (2.18a)$$

$$-\frac{d\tilde{I}(z)}{dz} = (G' + j\omega C') \tilde{V}(z). \quad (2.18b)$$

(相量形式的电报员方程)

2-4 传输线上的波传播

两个一阶耦合方程 (2.18a) 和 (2.18b) 可以组合起来得到两个二阶非耦合波动方程: 一个用于 $\tilde{V}(z)$, 另一个用于 $\tilde{I}(z)$ 。 $\tilde{V}(z)$ 的波动方程是通过首先对方程两边求导得出的。 (2.18a) 关于 z , 导致

$$-\frac{d^2\tilde{V}(z)}{dz^2} = (R' + j\omega L') \frac{d\tilde{I}(z)}{dz}. \quad (2.19)$$

然后, 将方程代入后。 (2.18b) 对于 $d\tilde{I}(z)/dz$, 方程(2.19)变为

$$\frac{d^2\tilde{V}(z)}{dz^2} - (R' + j\omega L')(G' + j\omega C') \tilde{V}(z) = 0, \quad (2.20)$$

或者

$$\frac{d^2\tilde{V}(z)}{dz^2} - \gamma^2 \tilde{V}(z) = 0, \quad (2.21)$$

(wave equation for $\tilde{V}(z)$)

在哪里

$$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}. \quad (2.22)$$

(propagation constant)

Application of the same steps to Eqs. (2.18a) and (2.18b) in reverse order leads to

$$\frac{d^2 \tilde{I}(z)}{dz^2} - \gamma^2 \tilde{I}(z) = 0. \quad (2.23)$$

(wave equation for $\tilde{I}(z)$)

The second-order differential equations (2.21) and (2.23) are called **wave equations** for $\tilde{V}(z)$ and $\tilde{I}(z)$, respectively, and γ is called the **complex propagation constant** of the transmission line. As such, γ consists of a real part α , called the **attenuation constant** of the line with units of Np/m, and an imaginary part β , called the **phase constant** of the line with units of rad/m. Thus,

$$\gamma = \alpha + j\beta \quad (2.24)$$

with

$$\alpha = \Re(\gamma) = \Re\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{Np/m}), \quad (2.25a)$$

(attenuation constant)

$$\beta = \Im(\gamma) = \Im\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{rad/m}). \quad (2.25b)$$

(phase constant)

2-4.1 Phasor-Domain Solution

In Eqs. (2.25a) and (2.25b), we choose the square-root solutions that give positive values for α and β . For passive transmission lines, α is either zero or positive. Most transmission lines, and all those considered in this chapter, are of the passive type. The gain region of a laser is an example of an active transmission line with a negative α .

The wave equations (2.21) and (2.23) have traveling wave solutions of the form:

$$\tilde{V}(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \quad (\text{V}), \quad (2.26a)$$

$$\tilde{I}(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \quad (\text{A}). \quad (2.26b)$$

As shown later, the $e^{-\gamma z}$ term represents a wave propagating in the $+z$ direction, while the $e^{\gamma z}$ term represents a wave propagating in the $-z$ direction (Fig. 2-9). Verification that these are indeed valid solutions is easily accomplished by substituting the expressions given by Eqs. (2.26a) and (2.26b), as well as their second derivatives, into Eqs. (2.21) and (2.23).

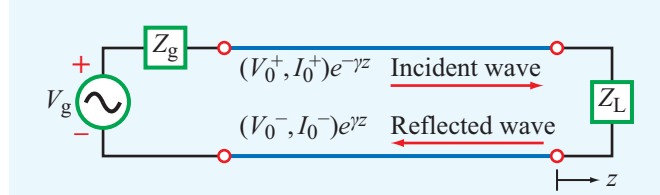


Figure 2-9 In general, a transmission line can support two traveling waves: an incident wave (with voltage and current amplitudes (V_0^+, I_0^+)) traveling along the $+z$ direction (towards the load) and a reflected wave (with (V_0^-, I_0^-)) traveling along the $-z$ direction (towards the source).

In their present form, the solutions given by Eqs. (2.26a) and (2.26b) contain four unknowns: the wave amplitudes (V_0^+, I_0^+) of the $+z$ propagating wave and (V_0^-, I_0^-) of the $-z$ propagating wave. We can easily relate the current wave amplitudes, I_0^+ and I_0^- , to the voltage wave amplitudes, V_0^+ and V_0^- , by using Eq. (2.26a) in Eq. (2.18a) and then solving for the current $\tilde{I}(z)$. The process leads to

$$\tilde{I}(z) = \frac{\gamma}{R' + j\omega L'} [V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}]. \quad (2.27)$$

Comparison of each term with the corresponding term in Eq. (2.26b) leads us to conclude that

$$\frac{V_0^+}{I_0^+} = Z_0 = \frac{-V_0^-}{I_0^-}, \quad (2.28)$$

where

$$Z_0 = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \quad (\Omega), \quad (2.29)$$

is called the **characteristic impedance** of the line.

► It should be noted that Z_0 is equal to the ratio of the voltage amplitude to the current amplitude for each of the traveling waves individually (with an additional minus sign in the case of the $-z$ propagating wave), but it is not equal to the ratio of the total voltage $\tilde{V}(z)$ to the total current $\tilde{I}(z)$ unless one of the two waves is absent. ◀

It seems reasonable that the voltage-to-current ratios of the two waves V_0^+/I_0^+ and V_0^-/I_0^- are both related to the same quantity, namely Z_0 , but it is not immediately obvious as

对等式应用相同的步骤。(2.18a) 和 (2.18b) 按相反顺序可得

$$\frac{d^2 \tilde{I}(z)}{dz^2} - \gamma^2 \tilde{I}(z) = 0. \quad (2.23)$$

(wave equation for $\tilde{I}(z)$)

二阶微分方程 (2.21) 和 (2.23) 分别称为 $\tilde{V}(z)$ 和 $\tilde{I}(z)$ 的波动方程, γ 称为传输线的复传播常数。因此, γ 由实部 α (称为线的衰减常数, 单位为 Np/m) 和虚部 β (称为线的相位常数, 单位为 rad/m) 组成。因此,

$$\gamma = \alpha + j\beta \quad (2.24)$$

和

$$\alpha = \Re(\gamma) = \Re\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{Np/m}), \quad (2.25a)$$

(attenuation constant)

$$\beta = \Im(\gamma) = \Im\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{rad/m}). \quad (2.25b)$$

(phase constant)

2-4.1 相量域解

在方程式中。(2.25a) 和 (2.25b) 中, 我们选择为 α 和 β 提供正值的平方根解。对于无源传输线, α 为零或正。大多数传输线以及本章中考虑的所有传输线都是无源类型。激光器的增益区域是具有负 α 的有源传输线的示例。

波动方程 (2.21) 和 (2.23) 具有以下形式的行波解:

$$\tilde{V}(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \quad (\text{V}), \quad (2.26a)$$

$$\tilde{I}(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \quad (\text{A}). \quad (2.26b)$$

如后所示, $e^{-\gamma z}$ 项表示沿 $+z$ 方向传播的波, 而 $e^{\gamma z}$ 项表示沿 $-z$ 方向传播的波 (图2-9)。通过替换方程式给出的表达式, 可以轻松验证这些确实是有效的解决方案。将 (2.26a) 和 (2.26b) 及其二阶数代入方程。(2.21) 和 (2.23)。

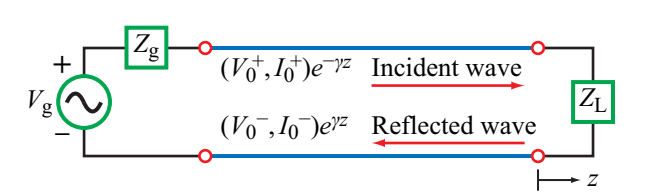


图 2-9 一般来说, 传输线可以支持两种行波: 沿 $+z$ 方向 (朝向负载) 传播的入射波 (电压和电流幅度为 (V_0^+, I_0^+)) 和沿 $-z$ 方向 (朝向源) 传播的反射波 (具有 (V_0^-, I_0^-))。

以目前的形式, 方程给出的解决方案。(2.26a)和(2.26b)包含四个未知数: $+z$ 传播波的波幅(V_0^+, I_0^+)和 $-z$ 传播波的(V_0^-, I_0^-)。通过使用等式, 我们可以轻松地将电流波幅度 I_0^+ 和 I_0^- 与电压波幅度 V_0^+ 和 V_0^- 联系起来。方程 (2.26a) (2.18a) 然后求解电流的 $\tilde{I}(z)$ 。该过程导致

$$\tilde{I}(z) = \frac{\gamma}{R' + j\omega L'} [V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}]. \quad (2.27)$$

每一项与等式中相应项的比较。(2.26b) 使我们得出结论

$$\frac{V_0^+}{I_0^+} = Z_0 = \frac{-V_0^-}{I_0^-}, \quad (2.28)$$

在哪里

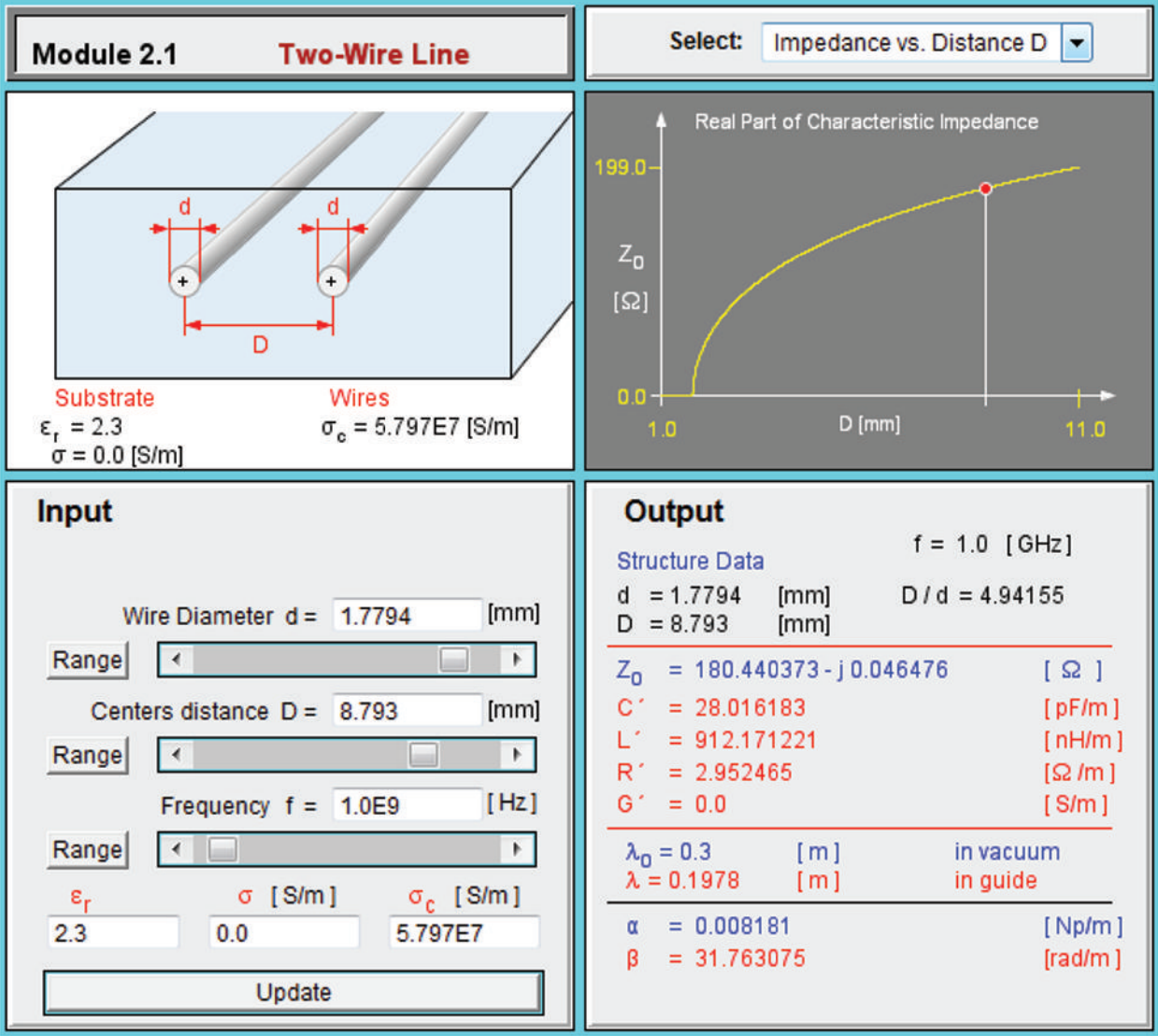
$$Z_0 = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \quad (\Omega), \quad (2.29)$$

称为线路的特性阻抗。

► 应该注意的是, Z_0 分别等于每个行波的电压幅度与电流幅度之比 (在 $-z$ 传播波的情况下带有附加负号), 但它不等于总电压 $\tilde{V}(z)$ 与总电流 $\tilde{I}(z)$ 的比率, 除非两个波之一不存在。◀

两个波 V_0^+/I_0^+ 和 V_0^-/I_0^- 的电压电流比都与相同的量 (即 Z_0) 相关, 这似乎是合理的, 但并不立即显而易见, 因为

Module 2.1 Two-Wire Line The input data specifies the geometric and electric parameters of a two-wire transmission line. The output includes the calculated values for the line parameters, characteristic impedance Z_0 , and attenuation and phase constants, as well as plots of Z_0 as a function of d and D .

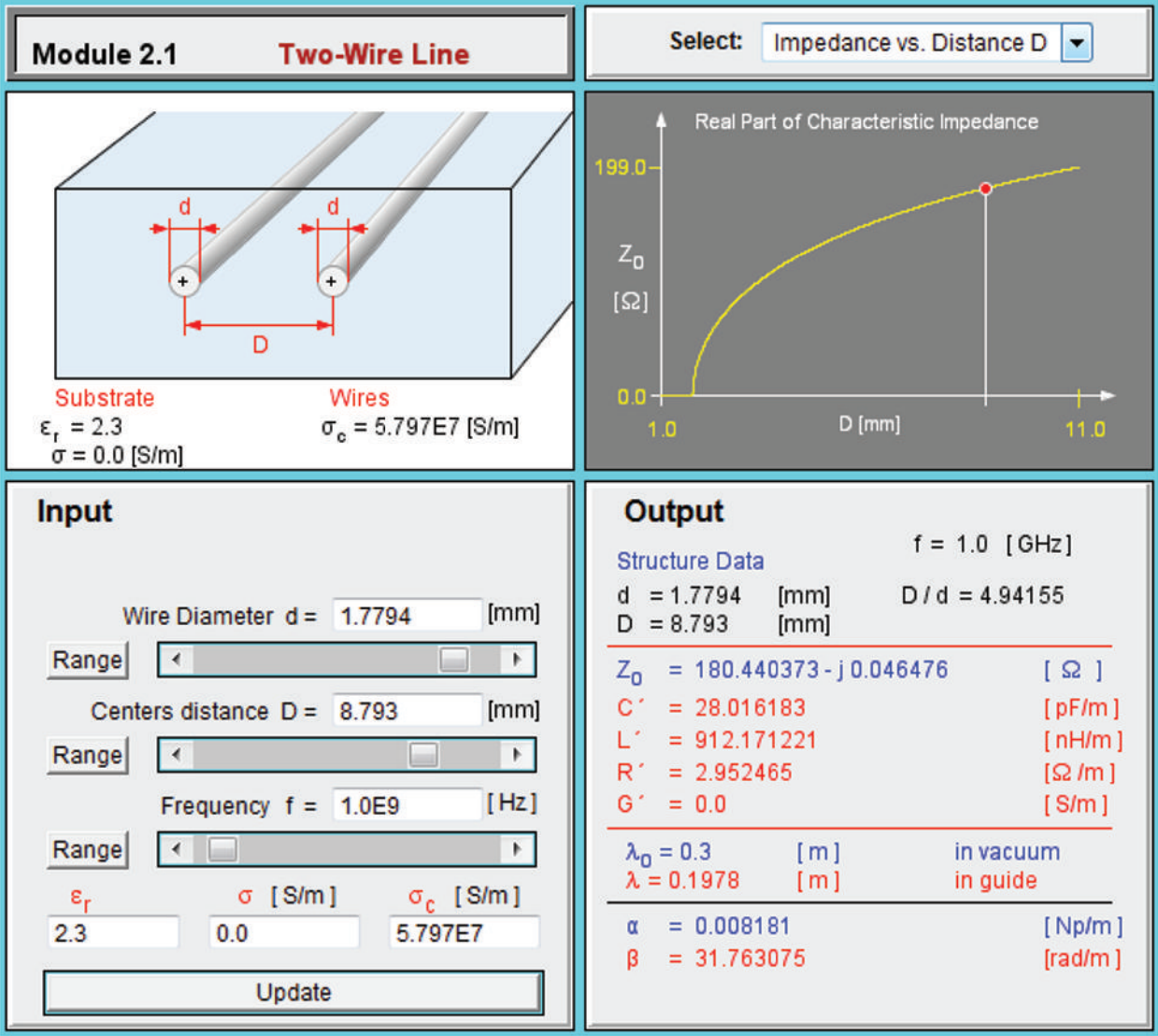


to why one of the ratios is the negative of the other. The explanation, which is available in more detail in Chapter 7, is based on a directional rule that specifies the relationships between the directions of the electric and magnetic fields of a TEM wave and its direction of propagation. On a transmission line, the voltage is related to the electric field $\mathbf{E}$ and the current is related to the magnetic field $\mathbf{H}$. To satisfy the

directional rule, reversing the direction of propagation requires reversal of the direction (or polarity) of I relative to V . Hence, $V_0^-/I_0^- = -V_0^+/I_0^+$. In terms of Z_0 , Eq. (2.27) can be cast in the form

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{\gamma z}. \quad (2.30)$$

模块 2.1 两线传输线 输入数据指定两线传输线的几何和电气参数。输出包括线路参数、特性阻抗 Z_0 、衰减和相位常数的计算值，以及作为 d 和 D 函数的 Z_0 绘图。

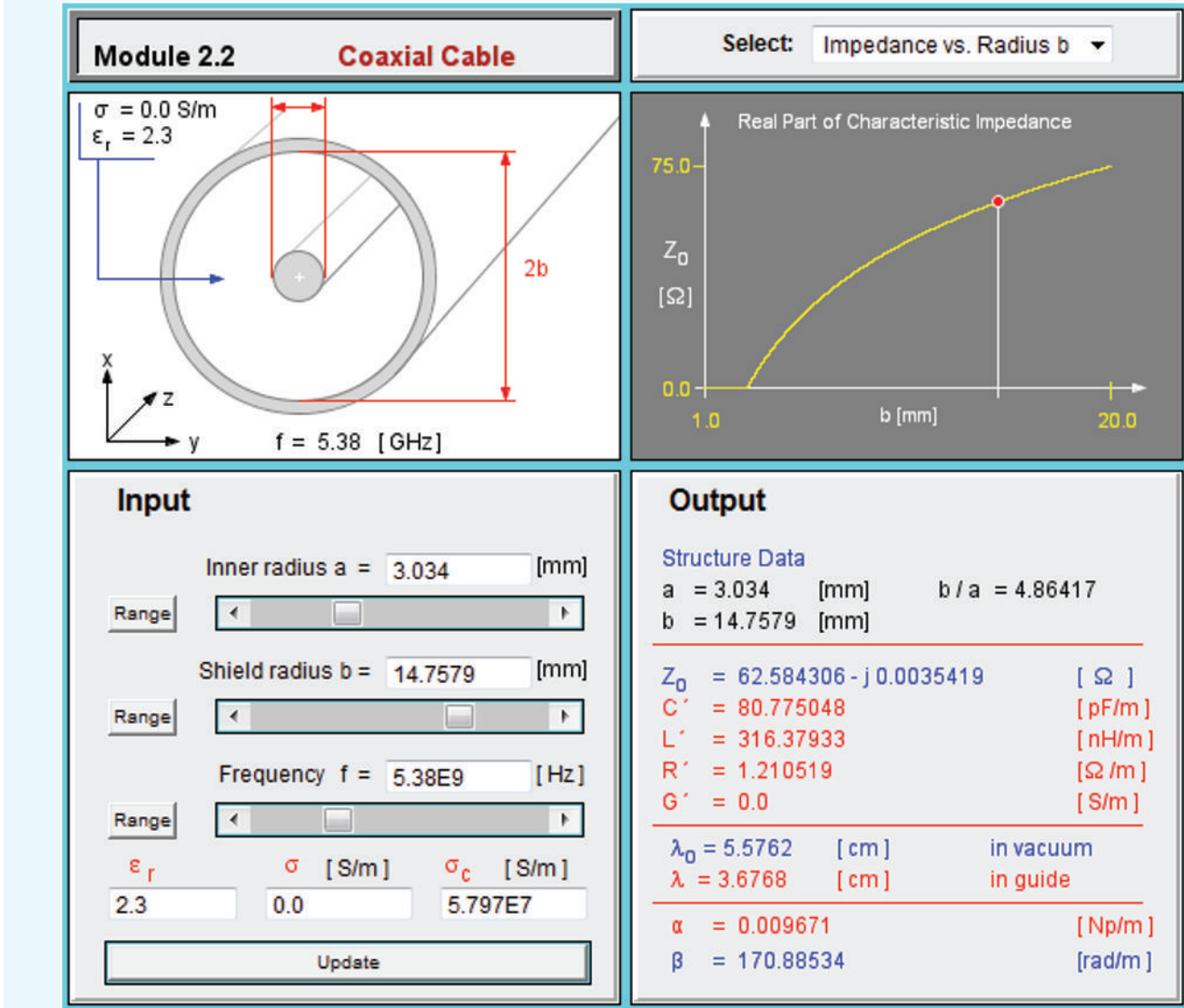


为什么其中一个比率是另一个比率的负数。第 7 章中有更详细的解释，它基于方向规则，该规则指定了 TEM 波的电场和磁场方向及其传播方向之间的关系。在传输线上，电压与电场 $\mathbf{E}$ 相关，电流与磁场 $\mathbf{H}$ 相关。满足

根据方向规则，反转传播方向需要反转 I 相对于 V 的方向（或极性）。因此， $V_0^-/I_0^- = -V_0^+/I_0^+$ 。就 Z_0 而言，方程：(2.27) 可以转换为以下形式

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{\gamma z}. \quad (2.30)$$

Module 2.2 Coaxial Line Except for changing the geometric parameters to those of a coaxial transmission line, this module offers the same output information as Module 2.1.



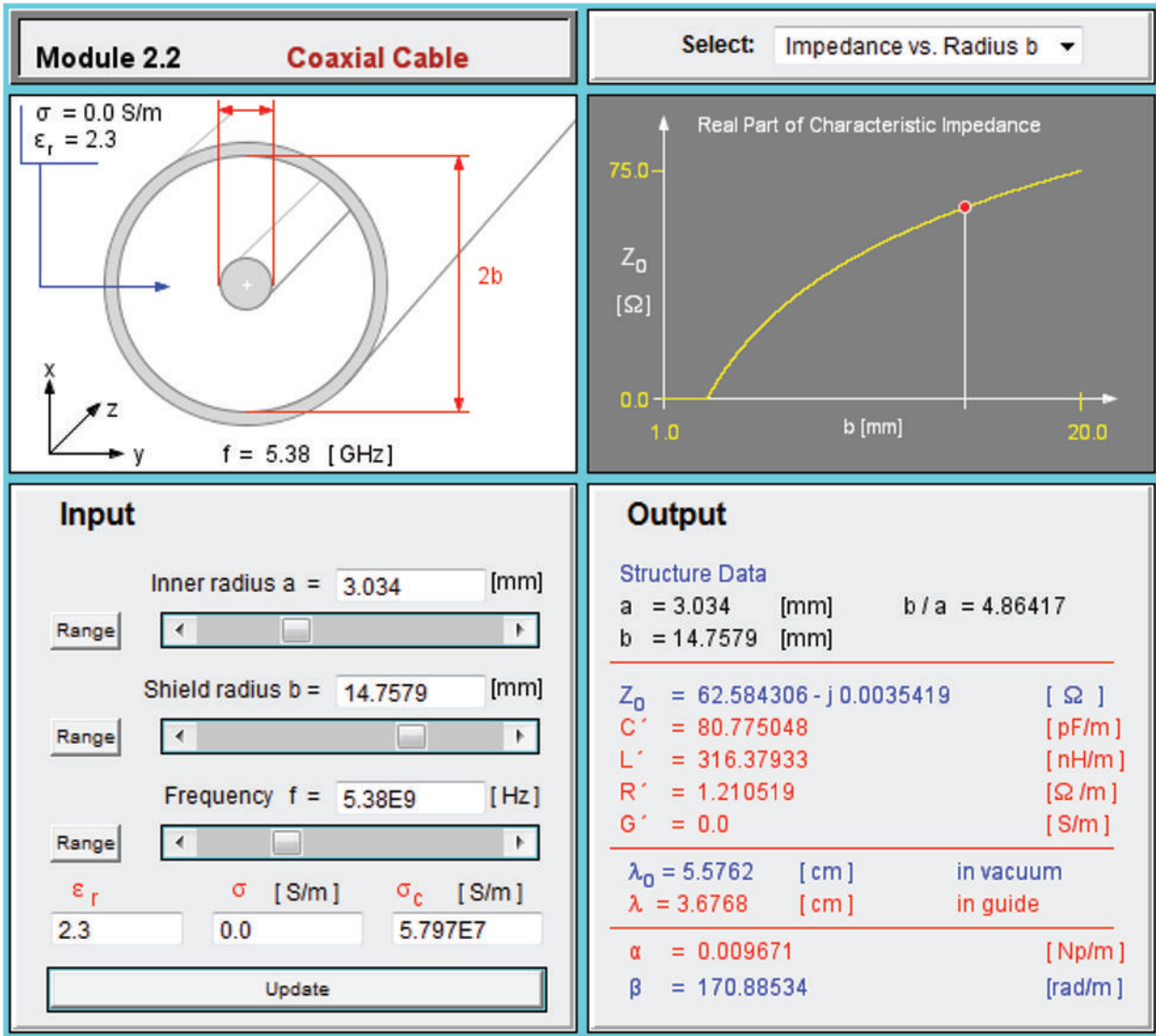
According to Eq. (2.29), the characteristic impedance Z_0 is determined by the angular frequency ω of the wave traveling along the line and the four line parameters (R' , L' , G' , and C'). These, in turn, are determined by the line geometry and its constitutive parameters. Consequently, the combination of Eqs. (2.26a) and (2.30) now contains only two unknowns, namely V_0^+ and V_0^- , as opposed to four.

2-4.2 Time-Domain Solution for $v(z, t)$

In later sections, we apply boundary conditions at the source and load ends of the transmission line to obtain expressions for the remaining wave amplitudes V_0^+ and V_0^- . In general, each is a complex quantity characterized by a magnitude and a phase angle:

$$V_0^+ = |V_0^+|e^{j\phi^+}, \quad (2.31a)$$

模块2.2 同轴线 除了将几何参数更改为同轴传输线的几何参数外，该模块提供与模块2.1 相同的输出信息。



根据方程。根据方程 (2.29)，特性阻抗 Z_0 由沿线传播的波的角频率 ω 和四个线参数 (R' , L' , G' 和 C') 确定。这些又由线的几何形状及其本构参数决定。因此，等式的组合。(2.26a) 和 (2.30) 现在仅包含两个未知数，即 V_0^+ 和 V_0^- ，而不是四个。

2-4.2 $v(z, t)$ 的时域解

在后面的部分中，我们在传输线的源端和负载端应用边界条件，以获得剩余波幅 V_0^+ 和 V_0^- 的表达式。一般来说，每个都是一个由幅度和相位角表征的复量：

$$V_0^+ = |V_0^+|e^{j\phi^+}, \quad (2.31a)$$

$$V_0^- = |V_0^-|e^{j\phi^-}. \quad (2.31b)$$

After substituting these definitions in Eq. (2.26a) and using Eq. (2.24) to decompose γ into its real and imaginary parts, we can convert back to the time domain to obtain an expression for $v(z, t)$, the instantaneous voltage on the line:

$$\begin{aligned} v(z, t) &= \Re\{\tilde{V}(z)e^{j\omega t}\} \\ &= \Re\left[(V_0^+e^{-\gamma z} + V_0^-e^{\gamma z})e^{j\omega t}\right] \\ &= \Re[|V_0^+|e^{j\phi^+}e^{j\omega t}e^{-(\alpha+j\beta)z} + |V_0^-|e^{j\phi^-}e^{j\omega t}e^{(\alpha+j\beta)z}]. \end{aligned}$$

The final expression for $v(z, t)$ is then given by

$$\begin{aligned} v(z, t) &= |V_0^+|e^{-\alpha z}\cos(\omega t - \beta z + \phi^+) \\ &\quad + |V_0^-|e^{\alpha z}\cos(\omega t + \beta z + \phi^-). \end{aligned} \quad (2.32)$$

From our review of waves in Section 1-4, we recognize the first term in Eq. (2.32) as a wave traveling in the $+z$ direction (the coefficients of t and z have opposite signs) and the second term as a wave traveling in the $-z$ direction (the coefficients of t and z are both positive). Both waves propagate with a phase velocity u_p given by Eq. (1.30):

$$u_p = f\lambda = \frac{\omega}{\beta}. \quad (2.33)$$

Because the wave is *guided* by the transmission line, λ often is called the ***guide wavelength***. The factor $e^{-\alpha z}$ accounts for the attenuation of the $+z$ propagating wave, and the factor $e^{\alpha z}$ accounts for the attenuation of the $-z$ propagating wave.

► The presence of two waves on the line propagating in opposite directions produces a ***standing wave***. ◀

To gain a physical understanding of what that means, we later provide a thorough treatment of standing waves in Section 2-6.2.

2-4.3 Time-Domain Solution for $i(z, t)$

In the phasor domain, the expression for $\tilde{I}(z)$ given by Eq. (2.30) is similar to that for $\tilde{V}(z)$, except that the current amplitudes are divided by the characteristic impedance of the transmission line, Z_0 , and the current wave traveling in the $-z$ direction has a minus sign (second term in Eq. (2.30)). Since Z_0 is, in general, a complex quantity (see Eq. (2.29)), we need to express it as

$$Z_0 = |Z_0|e^{j\phi_z}. \quad (2.34)$$

To obtain an expression for $i(z, t)$, we insert the expressions given by Eqs. (2.31) and (2.34) into Eq. (2.30), multiply both terms by $e^{j\omega t}$, and then take the real part:

$$\begin{aligned} i(z, t) &= \Re\{\tilde{I}(z)e^{j\omega t}\} = \Re\left[\frac{|V_0^+|}{|Z_0|}e^{j\phi^+}e^{-j\phi_z}e^{j\omega t}e^{-(\alpha+j\beta z)}\right. \\ &\quad \left.- \frac{|V_0^-|}{|Z_0|}e^{j\phi^-}e^{-j\phi_z}e^{j\omega t}e^{(\alpha+j\beta z)}\right], \end{aligned}$$

which yields

$$\begin{aligned} i(z, t) &= \frac{|V_0^+|}{|Z_0|}e^{-\alpha z}\cos(\omega t - \beta z + \phi^+ - \phi_z) \\ &\quad - \frac{|V_0^-|}{|Z_0|}e^{\alpha z}\cos(\omega t + \beta z + \phi^- - \phi_z). \end{aligned} \quad (2.35)$$

Comparison of the expression for $i(z, t)$ with that given by Eq. (2.32) for $v(z, t)$ reveals that:

- (a) The amplitudes of the current waves are reduced by $|Z_0|$ when compared with the amplitudes of the voltage waves.
- (b) The amplitude of the second term in Eq. (2.35)—corresponding to the current wave traveling in the negative z direction—has a minus sign.
- (c) The reference phase angle in the expression for $i(z, t)$ has an extra component, namely $-\phi_z$, when compared with the reference phase angle of $v(z, t)$.

Example 2-1: Air Line

An ***air line*** is a transmission line in which air separates the two conductors, which renders $G' = 0$ because $\sigma = 0$. In addition, assume that the conductors are made of a material with high conductivity so that $R' \approx 0$. For an air line with a characteristic impedance of $50\ \Omega$ and a phase constant of 20 rad/m at 700 MHz, find the line inductance L' and the line capacitance C' .

Solution: The following quantities are given:

$$\begin{aligned} Z_0 &= 50\ \Omega, \\ \beta &= 20\ \text{rad/m}, \\ f &= 700\ \text{MHz} = 7 \times 10^8\ \text{Hz}. \end{aligned}$$

With $R' = G' = 0$, Eqs. (2.25b) and (2.29) reduce to

$$\beta = \Im\left[\sqrt{(j\omega L')(j\omega C')}\right] = \Im\left(j\omega\sqrt{L'C'}\right) = \omega\sqrt{L'C'},$$

$$V_0^- = |V_0^-|e^{j\phi^-}. \quad (2.31b)$$

将这些定义代入等式后： (2.26a) 并使用方程。 (2.24) 将 γ 分解为实部和虚部，我们可以转换回时域，得到 $v(z, t)$ 的表达式，即线路上的瞬时电压：

$$\begin{aligned} v(z, t) &= \Re\{\tilde{V}(z)e^{j\omega t}\} \\ &= \Re\left[(V_0^+e^{-\gamma z} + V_0^-e^{\gamma z})e^{j\omega t}\right] \\ &= \Re[|V_0^+|e^{j\phi^+}e^{j\omega t}e^{-(\alpha+j\beta)z} + |V_0^-|e^{j\phi^-}e^{j\omega t}e^{(\alpha+j\beta)z}]. \end{aligned}$$

$v(z, t)$ 的最终表达式由下式给出

$$\begin{aligned} v(z, t) &= |V_0^+|e^{-\alpha z}\cos(\omega t - \beta z + \phi^+) \\ &\quad + |V_0^-|e^{\alpha z}\cos(\omega t + \beta z + \phi^-). \end{aligned} \quad (2.32)$$

从我们对第 1-4 节中的波的回顾中，我们认识到方程中的第一项。(2.32) 为沿 $+z$ 方向传播的波（ t 和 z 的系数具有相反的符号），第二项为沿 $-z$ 方向传播的波（ t 和 z 的系数均为正）。两个波都以等式给出的相速度 u_p 传播。（1.30）：

$$u_p = f\lambda = \frac{\omega}{\beta}. \quad (2.33)$$

由于波是由传输线引导的，因此 λ 通常称为波导波长。因子 $e^{-\alpha z}$ 说明了 $+z$ 传播波的衰减，因子 $e^{\alpha z}$ 说明了 $-z$ 传播波的衰减。

► 线上存在沿相反方向传播的两个波会产生驻波。 ◀

为了从物理上理解这意味着什么，我们稍后在 2-6.2 节中对驻波进行彻底的处理。

2-4.3 i(z,t)的时域解

在相量域中， $\tilde{I}(z)$ 的表达式由式 (1) 给出。方程 (2.30) 与 $\tilde{V}(z)$ 类似，不同之处在于电流幅值除以传输线的特性阻抗 Z_0 ，并且沿 $-z$ 方向传播的电流波具有负号（方程 (2.30) 中的第二项）。由于 Z_0 通常是一个复数（参见方程 (2.29)），我们需要将其表示为

$$Z_0 = |Z_0|e^{j\phi_z}. \quad (2.34)$$

为了获得 $i(z, t)$ 的表达式，我们插入方程 1 给出的表达式。将 (2.31) 和 (2.34) 代入方程。 (2.30)，将两项乘以 $e^{j\omega t}$ ，然后取实部：

$$\begin{aligned} i(z, t) &= \Re\{\tilde{I}(z)e^{j\omega t}\} = \Re\left[\frac{|V_0^+|}{|Z_0|}e^{j\phi^+}e^{-j\phi_z}e^{j\omega t}e^{-(\alpha+j\beta z)}\right. \\ &\quad \left.- \frac{|V_0^-|}{|Z_0|}e^{j\phi^-}e^{-j\phi_z}e^{j\omega t}e^{(\alpha+j\beta z)}\right], \end{aligned}$$

这产生

$$\begin{aligned} i(z, t) &= \frac{|V_0^+|}{|Z_0|}e^{-\alpha z}\cos(\omega t - \beta z + \phi^+ - \phi_z) \\ &\quad - \frac{|V_0^-|}{|Z_0|}e^{\alpha z}\cos(\omega t + \beta z + \phi^- - \phi_z). \end{aligned} \quad (2.35)$$

$i(z, t)$ 的表达式与方程给出的表达式的比较。(2.32) 对于 $v(z, t)$ 表明：

- (a) 与电压波的振幅相比，电流波的振幅减少了 $|Z_0|$ 。
- (b) 等式中第二项的幅度。(2.35)——对应于在负 z 方向上传播的电流波——有一个负号。
- (c) 与 $v(z, t)$ 的参考相位角相比， $i(z, t)$ 表达式中的参考相位角多了一个分量，即 $-\phi_z$ 。

示例 2-1: 航空线路

空气线路是一种传输线路，其中空气将两个导体分开，这使得 $G' = 0$ 因为 $\sigma = 0$ 。此外，假设导体由高电导率材料制成，因此 $R' \approx 0$ 。对于特性阻抗为 $50\ \Omega$ 且在 700 MHz 时相位常数为 20 rad/m 的空气线路，求出线路电感 L' 和线路电感电容 C' 。

解：给出下列数量：

$$\begin{aligned} Z_0 &= 50\ \Omega, \\ \beta &= 20\ \text{rad/m}, \\ f &= 700\ \text{MHz} = 7 \times 10^8\ \text{Hz}. \end{aligned}$$

当 $R' = G' = 0$ 时，方程(2.25b)和 (2.29) 减少为

$$\beta = \Im\left[\sqrt{(j\omega L')(j\omega C')}\right] = \Im\left(j\omega\sqrt{L'C'}\right) = \omega\sqrt{L'C'},$$

$$Z_0 = \sqrt{\frac{j\omega L'}{j\omega C'}} = \sqrt{\frac{L'}{C'}}$$

The ratio of β to Z_0 is $\beta/Z_0 = \omega C'$, or

$$C' = \frac{\beta}{\omega Z_0} = \frac{20}{2\pi \times 7 \times 10^8 \times 50} = 9.09 \times 10^{-11} \text{ (F/m)} \\ = 90.9 \text{ (pF/m)}.$$

From $Z_0 = \sqrt{L'/C'}$, it follows that

$$L' = Z_0^2 C' = (50)^2 \times 90.9 \times 10^{-12} = 2.27 \times 10^{-7} \text{ (H/m)} \\ = 227 \text{ (nH/m)}.$$

Exercise 2-3: Verify that Eq. (2.26a) indeed provides a solution to the wave equation (2.21).

Answer: (See )

Exercise 2-4: A two-wire air line has the following line parameters: $R' = 0.404 \text{ (m}\Omega/\text{m)}$, $L' = 2.0 \text{ (}\mu\text{H/m)}$, $G' = 0$, and $C' = 5.56 \text{ (pF/m)}$. For operation at 5 kHz, determine (a) the attenuation constant α , (b) the phase constant β , (c) the phase velocity u_p , and (d) the characteristic impedance Z_0 . (See )

Answer: (a) $\alpha = 3.37 \times 10^{-7} \text{ (Np/m)}$, (b) $\beta = 1.05 \times 10^{-4} \text{ (rad/m)}$, (c) $u_p = 3.0 \times 10^8 \text{ (m/s)}$, (d) $Z_0 = (600 - j1.9) \Omega = 600\angle-0.18^\circ \Omega$.

2-5 The Lossless Microstrip Line

Because its geometry is well suited for fabrication on printed circuit boards, the microstrip line is the most common interconnect configuration used in RF and microwave circuits. It consists of a narrow, very thin strip of copper (or another good conductor) printed on a dielectric substrate overlaying a ground plane (Fig. 2-10(a)). The presence of charges of opposite polarity on its two conducting surfaces gives rise to electric field lines between them (Fig. 2-10(b)). Also, the flow of current through the conductors (when part of a closed circuit) generates magnetic field loops around them, as illustrated in Fig. 2-10(b) for the narrow strip. Even though the patterns of **E** and **B** are not everywhere perfectly orthogonal, they are approximately so in the region between the conductors, which is where the **E** and **B** fields are concentrated the most. Accordingly, the microstrip line is considered a *quasi-TEM* transmission line, which allows us to describe its voltages and currents in terms of the one-dimensional TEM model of Section 2-4, namely Eqs. (2.26) through (2.33).

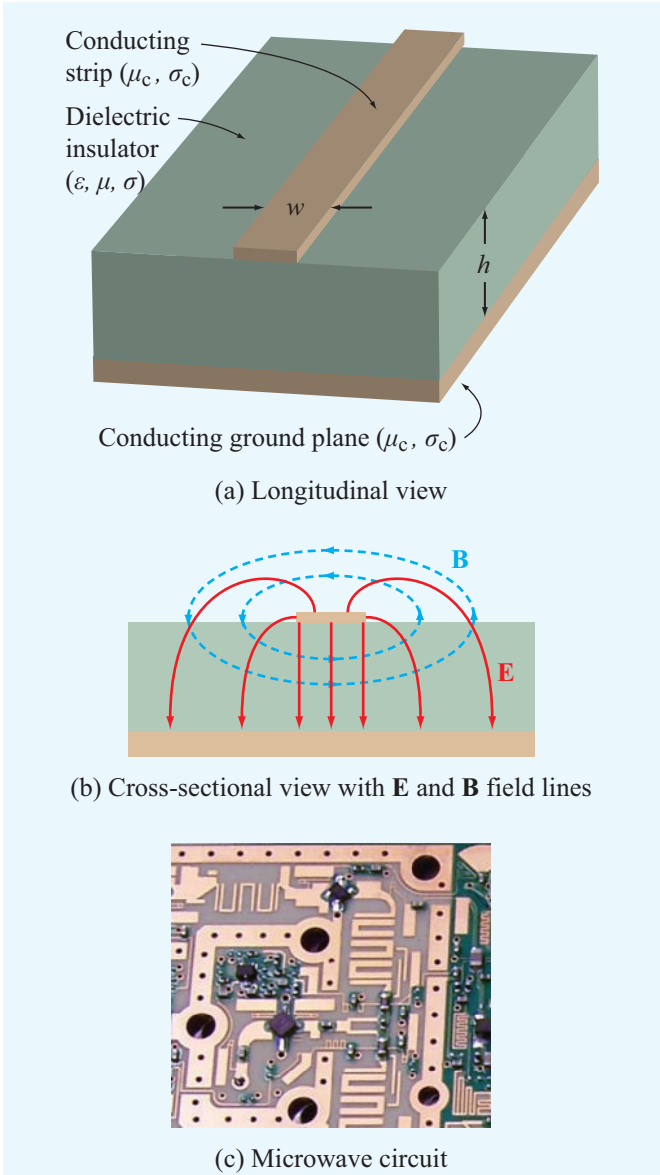


Figure 2-10 Microstrip line: (a) longitudinal view, (b) cross-sectional view, and (c) circuit example. (Courtesy of Prof. Gabriel Rebeiz, U. California at San Diego.)

The microstrip line has two geometric parameters, the *width of the elevated strip*, w , and the *thickness (height) of the dielectric layer*, h . We will ignore the thickness of the conducting strip because it has a negligible influence on the propagation properties of the microstrip line so long as the strip thickness is much smaller than the width w , which is almost

$$Z_0 = \sqrt{\frac{j\omega L'}{j\omega C'}} = \sqrt{\frac{L'}{C'}}$$

β 与 Z_0 的比率为 $\beta/Z_0 = \omega C'$, 或

$$C' = \frac{\beta}{\omega Z_0} = \frac{20}{2\pi \times 7 \times 10^8 \times 50} = 9.09 \times 10^{-11} \text{ (F/m)} \\ = 90.9 \text{ (pF/m)}.$$

从 $Z_0 = \sqrt{L'/C'}$ 可以得出

$$L' = Z_0^2 C' = (50)^2 \times 90.9 \times 10^{-12} = 2.27 \times 10^{-7} \text{ (H/m)} \\ = 227 \text{ (nH/m)}.$$

练习 2-3: 验证等式。 (2.26a) 确实提供了波动方程 (2.21) 的解。

答案: (参见 )

练习 2-4: 两线空气管路具有以下管路参数: $R' = 0.404 \text{ (m}\Omega/\text{m)}$ 、 $L' = 2.0 \text{ (}\mu\text{H/m)}$ 、 $G' = 0$ 和 $C' = 5.56 \text{ (pF/m)}$ 。对于工作频率为 5 kHz 的情况, 确定 (a) 衰减常数 α 、(b) 相位常数 β 、(c) 相速度 u_p 和 (d) 特征阻抗 Z_0 。 (参见 )

答案: (a) $\alpha = 3.37 \times 10^{-7} \text{ (Np/m)}$, (b) $\beta = 1.05 \times 10^{-4} \text{ (rad/m)}$, (c) $u_p = 3.0 \times 10^8 \text{ (m/s)}$, (d) $Z_0 = (600 - j1.9) \Omega = 600\angle-0.18^\circ \Omega$ 。

2-5 无损微带线

由于其几何形状非常适合在印刷电路板上制造, 因此微带线是射频和微波电路中最常用的互连配置。它由一条窄且非常薄的铜条 (或其他良导体) 组成, 印刷在覆盖接地层的电介质基板上 (图 2-10(a))。其两个导电表面上存在相反极性的电荷, 从而在它们之间产生电场线 (图2-10 (b))。此外, 流经导体 (作为闭合电路的一部分) 的电流会在导体周围产生磁场环, 如图 2-10(b) 中的窄带所示。尽管 **E** 和 **B** 的图案并非处处都完全正交, 但它们在导体之间的区域大致如此, 这是 **E** 和 **B** 场最集中的地方。因此, 微带线被认为是准 TEM 传输线, 这使我们能够根据第 2-4 节的一维 TEM 模型 (即方程 1) 来描述其电压和电流。 (2.26) 到 (2.33)。

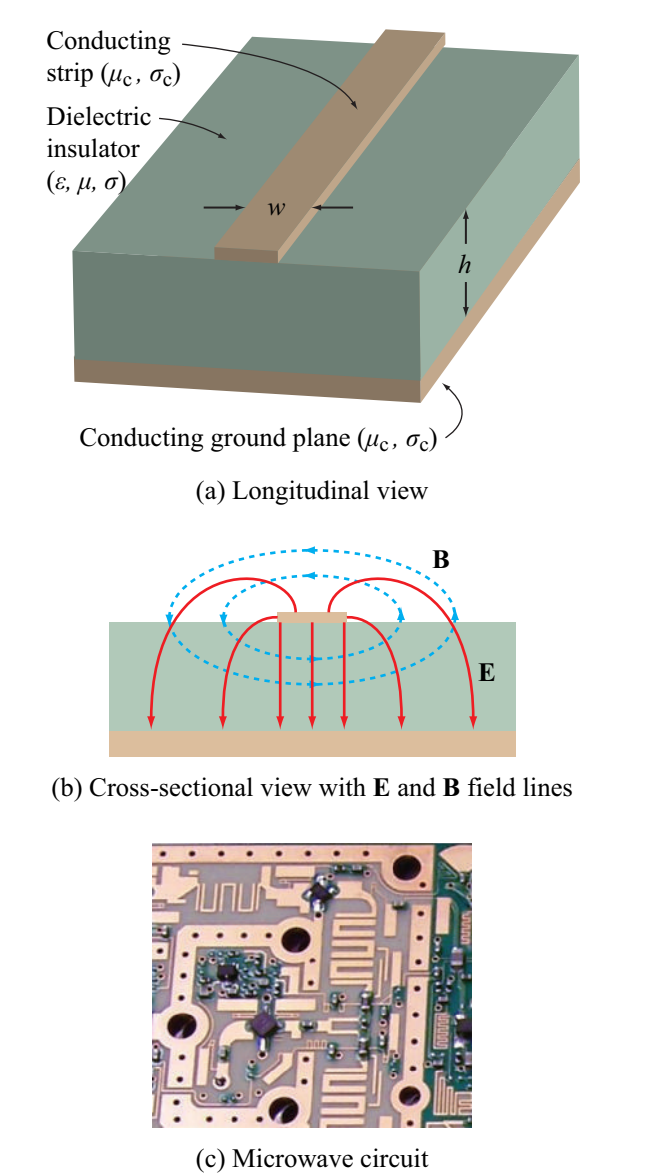


图 2-10 微带线: (a) 纵向视图, (b) 横截面视图, (c) 电路示例。 (由美国加州圣地亚哥的 Gabriel Rebeiz 教授提供。)

微带线有两个几何参数, 即抬高带的宽度 w 和介电层的厚度 (高度) h 。我们将忽略导电带的厚度, 因为只要导电带的厚度远小于宽度 w , 它对微带线的传播特性的影响就可以忽略不计, 宽度 w 几乎是

always the case in practice. Also, we assume the substrate material to be a perfect dielectric with $\sigma = 0$ and the metal strip and ground plane to be perfect conductors with $\sigma_c \approx \infty$. These two assumptions simplify the analysis considerably without incurring significant error. Finally, we set $\mu = \mu_0$, which is always true for the dielectric materials used in the fabrication of microstrip lines. These simplifications reduce the number of geometric and material parameters to three, namely w , h , and ϵ .

Electric field lines always start on the conductor carrying positive charges and end on the conductor carrying negative charges. For the coaxial, two-wire, and parallel-plate lines shown in the upper part of **Fig. 2-4**, the field lines are confined to the region between the conductors. A characteristic attribute of such transmission lines is that the phase velocity of a wave traveling along any one of them is given by

$$u_p = \frac{c}{\sqrt{\epsilon_r}}, \quad (2.36a)$$

where c is the velocity of light in free space and ϵ_r is the relative permittivity of the dielectric medium between the conductors.

2-5.1 Effective Permittivity

In the microstrip line, even though most of the electric field lines connecting the strip to the ground plane do pass directly through the dielectric substrate, a few go through both the air region above the strip and the dielectric layer (**Fig. 2-10(b)**). This nonuniform mixture can be accounted for by defining an **effective relative permittivity** ϵ_{eff} such that the phase velocity is given by an expression that resembles Eq. (2.36a), namely

$$u_p = \frac{c}{\sqrt{\epsilon_{\text{eff}}}}. \quad (2.36b)$$

Methods for calculating the propagation properties of the microstrip line are quite complicated and beyond the scope of this text. However, it is possible to use curve-fit approximations to rigorous solutions to arrive at the following set of expressions:*

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \left(\frac{\epsilon_r - 1}{2} \right) \left(1 + \frac{10}{s} \right)^{-xy}, \quad (2.37a)$$

where s is the **width-to-thickness ratio**,

$$s = \frac{w}{h}, \quad (2.37b)$$

*D. H. Schrader, *Microstrip Circuit Analysis*, Prentice Hall, 1995, pp. 31–32.

and x and y are intermediate variables given by

$$x = 0.56 \left[\frac{\epsilon_r - 0.9}{\epsilon_r + 3} \right]^{0.05}, \quad (2.38a)$$

$$y = 1 + 0.02 \ln \left(\frac{s^4 + 3.7 \times 10^{-4} s^2}{s^4 + 0.43} \right) + 0.05 \ln(1 + 1.7 \times 10^{-4} s^3). \quad (2.38b)$$

2-5.2 Characteristic Impedance

The characteristic impedance of the microstrip line is given by

$$Z_0 = \frac{60}{\sqrt{\epsilon_{\text{eff}}}} \ln \left\{ \frac{6 + (2\pi - 6)e^{-t}}{s} + \sqrt{1 + \frac{4}{s^2}} \right\}, \quad (2.39)$$

with

$$t = \left(\frac{30.67}{s} \right)^{0.75}. \quad (2.40)$$

Figure 2-11 displays plots of Z_0 as a function of s for various types of dielectric materials.

The corresponding line and propagation parameters are given by

$$R' = 0 \quad (\text{because } \sigma_c = \infty), \quad (2.41a)$$

$$G' = 0 \quad (\text{because } \sigma = 0), \quad (2.41b)$$

$$C' = \frac{\sqrt{\epsilon_{\text{eff}}}}{Z_0 c}, \quad (2.41c)$$

$$L' = Z_0^2 C', \quad (2.41d)$$

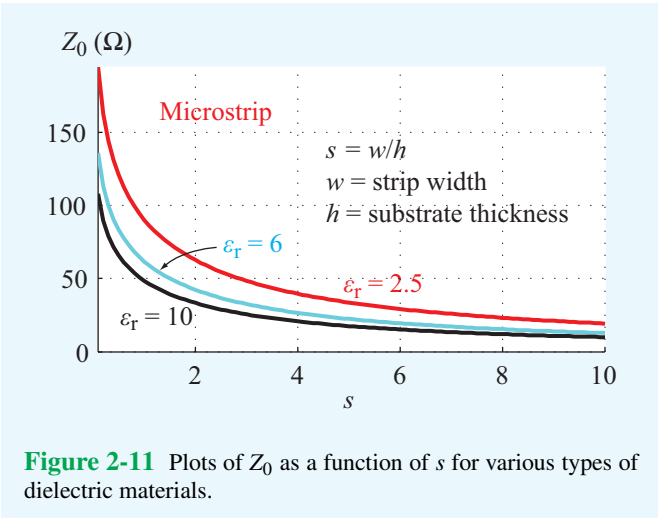


Figure 2-11 Plots of Z_0 as a function of s for various types of dielectric materials.

实践中总是如此。此外，我们假设基板材料是 $\sigma = 0$ 的理想电介质，金属条和接地平面是 $\sigma_c \approx \infty$ 的理想导体。这两个假设大大简化了分析，而不会产生重大错误。最后，我们设置 $\mu = \mu_0$ ，这对于微带线制造中使用的介电材料始终成立。这些简化将几何和材料参数的数量减少到三个，即 w 、 h 和 ϵ 。

电场线总是从带正电荷的导体开始，到带负电荷的导体结束。对于图 2-4 上半部分所示的同轴、双线和平行板线，场线被限制在导体之间的区域。这种传输线的一个特征是沿着其中任何一条传播的波的相速度由下式给出

$$u_p = \frac{c}{\sqrt{\epsilon_r}}, \quad (2.36a)$$

其中 c 是自由空间中的光速， ϵ_r 是导体之间介电介质的相对介电常数。

2-5.1 有效介电常数

在微带线中，尽管大多数将带状线连接到地平面的电场线直接穿过介电基板，但也有少数电场线同时穿过带状线上方的空气区域和介电层（图2-10（b））。这种不均匀的混合物可以通过定义有效相对介电常数 ϵ_{eff} 来解释，这样相速度就可以由类似于方程 1 的表达式给出。 (2.36a), 即

$$u_p = \frac{c}{\sqrt{\epsilon_{\text{eff}}}}. \quad (2.36b)$$

计算微带线传播特性的方法相当复杂，超出了本文的范围。然而，可以使用严格解的曲线拟合近似来得到以下表达式集：*

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \left(\frac{\epsilon_r - 1}{2} \right) \left(1 + \frac{10}{s} \right)^{-xy}, \quad (2.37a)$$

其中 s 是宽厚比，

$$s = \frac{w}{h}, \quad (2.37b)$$

*D. H. Schrader, *Microstrip Circuit Analysis*, Prentice Hall, 1995, pp. 31–32.

x 和 y 是由下式给出的中间变量

$$x = 0.56 \left[\frac{\epsilon_r - 0.9}{\epsilon_r + 3} \right]^{0.05}, \quad (2.38a)$$

$$y = 1 + 0.02 \ln \left(\frac{s^4 + 3.7 \times 10^{-4} s^2}{s^4 + 0.43} \right) + 0.05 \ln(1 + 1.7 \times 10^{-4} s^3). \quad (2.38b)$$

2-5.2 特性阻抗

微带线的特性阻抗由 b 给出

$$Z_0 = \frac{60}{\sqrt{\epsilon_{\text{eff}}}} \ln \left\{ \frac{6 + (2\pi - 6)e^{-t}}{s} + \sqrt{1 + \frac{4}{s^2}} \right\}, \quad (2.39)$$

和

$$t = \left(\frac{30.67}{s} \right)^{0.75}. \quad (2.40)$$

图 2-11 显示了各种类型介电材料的 Z_0 与 s 的函数关系图。

相应的线路和传播参数由下式给出

$$R' = 0 \quad (\text{because } \sigma_c = \infty), \quad (2.41a)$$

$$G' = 0 \quad (\text{because } \sigma = 0), \quad (2.41b)$$

$$C' = \frac{\sqrt{\epsilon_{\text{eff}}}}{Z_0 c}, \quad (2.41c)$$

$$L' = Z_0^2 C', \quad (2.41d)$$

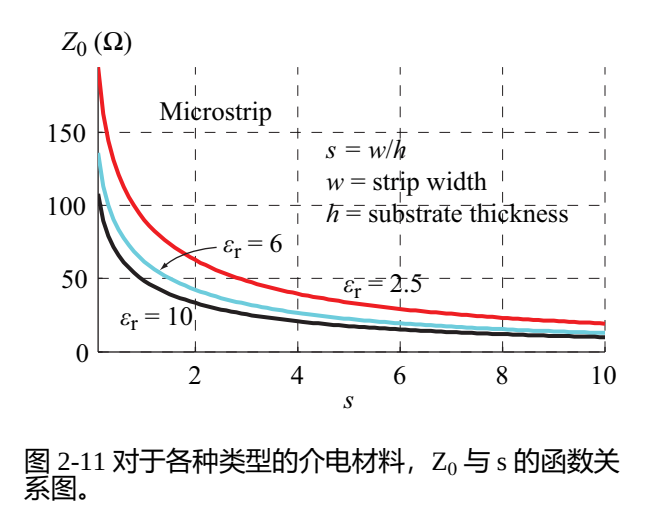
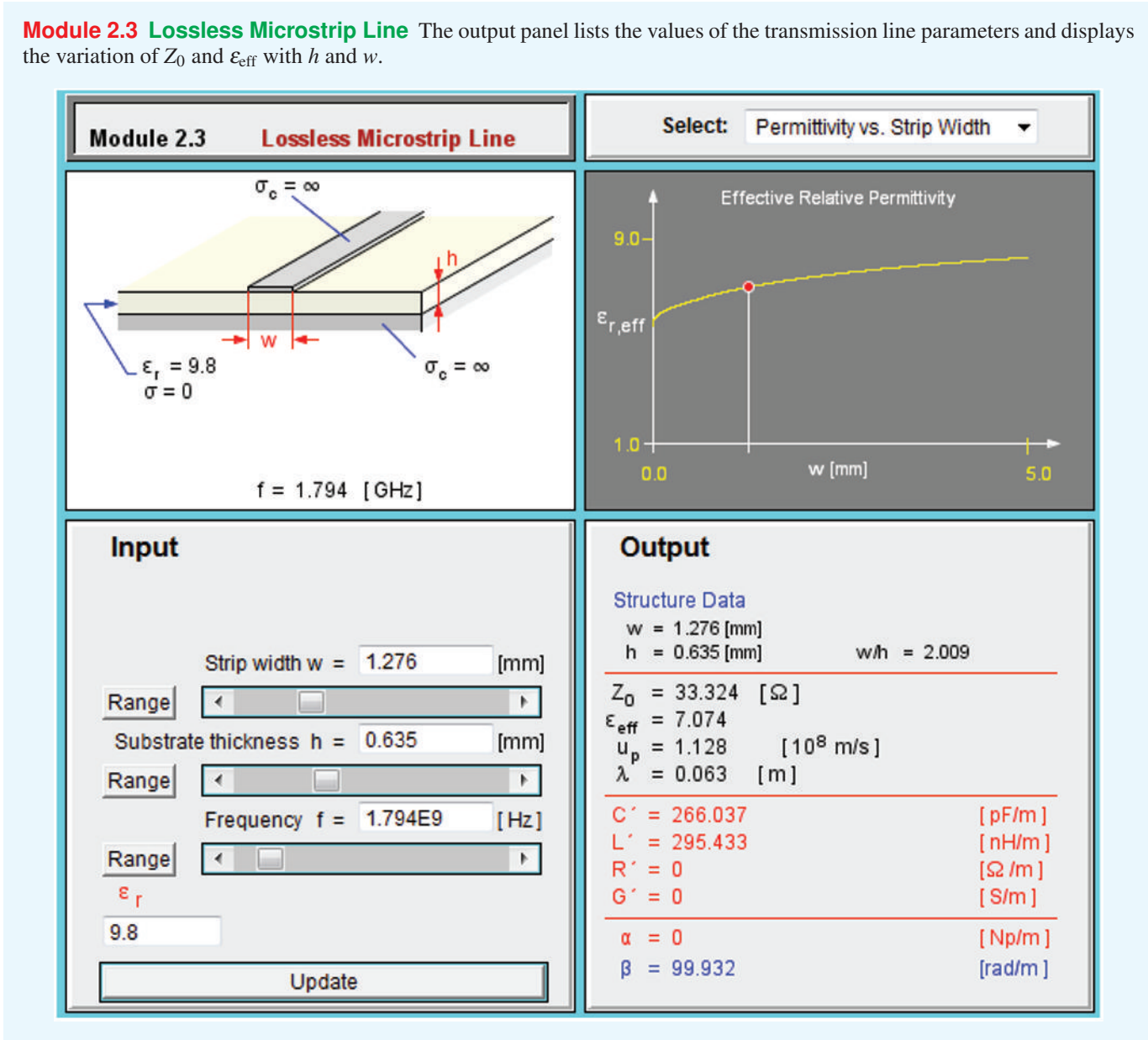
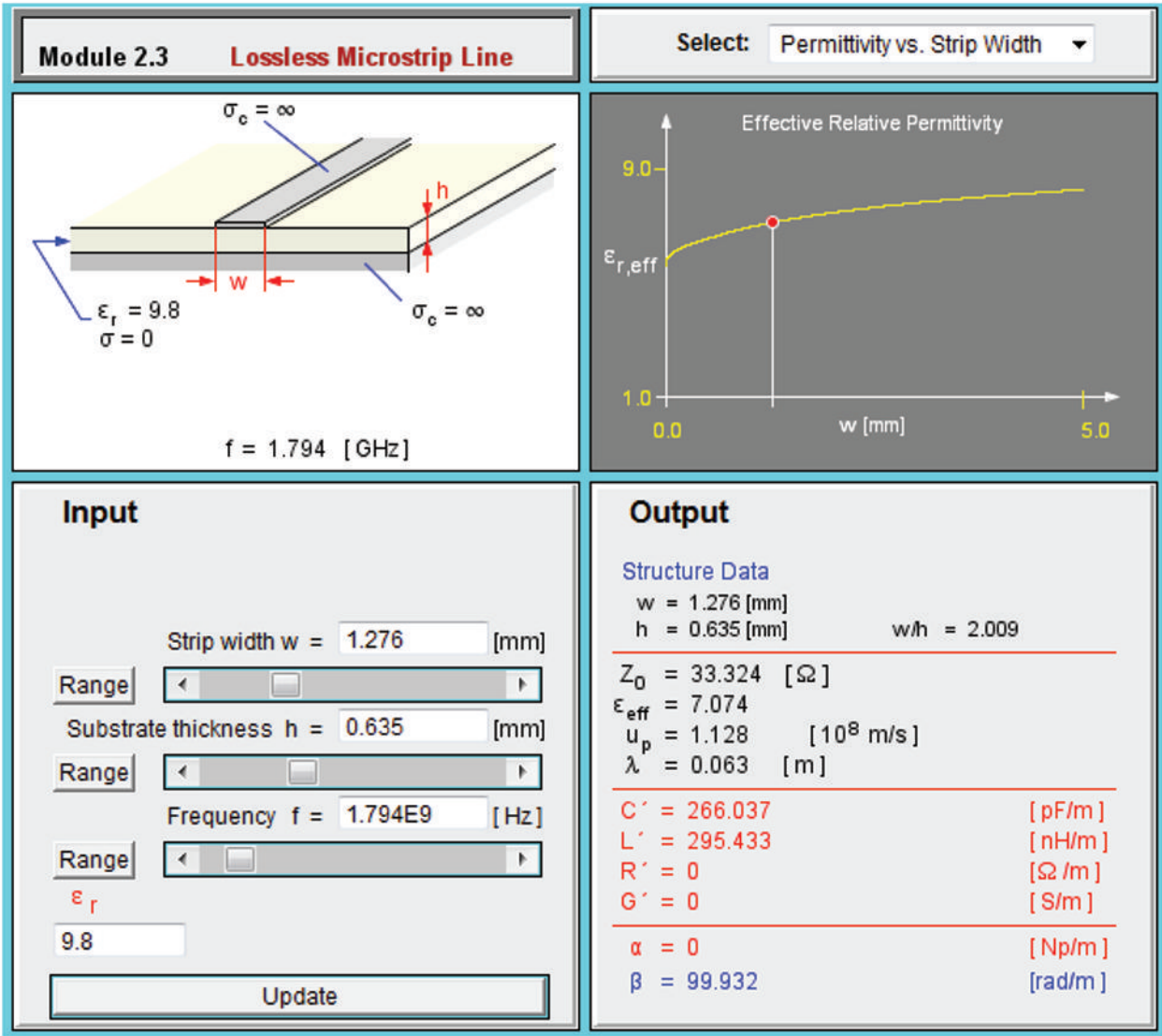


图 2-11 对于各种类型的介电材料， Z_0 与 s 的函数关系图。



模块 2.3 无损微带线 输出面板列出了传输线参数的值，并显示 Z_0 和 ϵ_{eff} 随 h 和 w 的变化。



$$\alpha = 0 \quad (\text{because } R' = G' = 0), \quad (2.41e)$$
$$\beta = \frac{\omega}{c} \sqrt{\epsilon_{\text{eff}}}. \quad (2.41f)$$

2-5.3 Design Process

The preceding expressions allow us to compute the values of Z_0 and the other propagation parameters when given values for ϵ_r , h , and w . This is exactly what is needed in order to

analyze a circuit containing a microstrip transmission line. To perform the reverse process, namely to design a microstrip line by selecting values for its w and h such that their ratio yields the required value of Z_0 (to satisfy design specifications), we need to express s in terms of Z_0 . The expression for Z_0 given by Eq. (2.39) is rather complicated, so inverting it to obtain an expression for s in terms of Z_0 is rather difficult. An alternative option is to generate a family of curves similar to those displayed in Fig. 2-11 and to use them to estimate

$$\alpha = 0 \quad (\text{because } R' = G' = 0), \quad (2.41e)$$

$$\beta = \frac{\omega}{c} \sqrt{\epsilon_{\text{eff}}}. \quad (2.41f)$$

2-5.3 设计流程

当给定 ϵ_r 、 h 和 w 的值时，前面的表达式允许我们计算 Z_0 和其他传播参数的值。这正是所需要的

分析包含微带传输线的电路。为了执行相反的过程，即通过选择 w 和 h 的值来设计微带线，使得它们的比率产生所需的 Z_0 （值以满足设计规范），我们需要用 Z_0 来表达 s 。 Z_0 的表达式由等式给出。(2.39) 相当复杂，因此将其求逆以获得 s 用 Z_0 表示的表达式相当困难。另一种选择是生成类似于图 2-11 中显示的曲线族，并使用它们来估计

s for a specified value of Z_0 . A logical extension of the graphical approach is to generate curve-fit expressions that provide high-accuracy estimates of s . The error associated with the following formulas is less than 2%:

(a) For $Z_0 \leq (44 - 2\epsilon_r) \, \Omega$,

$$s = \frac{w}{h} = \frac{2}{\pi} \left\{ (q - 1) - \ln(2q - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left[\ln(q - 1) + 0.29 - \frac{0.52}{\epsilon_r} \right] \right\} \quad (2.42)$$

with

$$q = \frac{60\pi^2}{Z_0\sqrt{\epsilon_r}},$$

and

(b) for $Z_0 \geq (44 - 2\epsilon_r) \, \Omega$,

$$s = \frac{w}{h} = \frac{8e^p}{e^{2p} - 2}, \quad (2.43a)$$

with

$$p = \sqrt{\frac{\epsilon_r + 1}{2}} \frac{Z_0}{60} + \left(\frac{\epsilon_r - 1}{\epsilon_r + 1} \right) \left(0.23 + \frac{0.12}{\epsilon_r} \right). \quad (2.43b)$$

The above expressions presume that ϵ_r , the relative permittivity of the dielectric substrate, has already been specified. For typical substrate materials including Duroid, Teflon, silicon, and sapphire, ϵ_r ranges between 2 and 15.

Example 2-2: Microstrip Line

A $50 \, \Omega$ microstrip line uses a 0.5-mm thick sapphire substrate with $\epsilon_r = 9$. What is the width of its copper strip?

Solution: Since $Z_0 = 50 > 44 - 18 = 32$, we should use Eq. (2.43):

$$\begin{aligned} p &= \sqrt{\frac{\epsilon_r + 1}{2}} \times \frac{Z_0}{60} + \left(\frac{\epsilon_r - 1}{\epsilon_r + 1} \right) \left(0.23 + \frac{0.12}{\epsilon_r} \right) \\ &= \sqrt{\frac{9 + 1}{2}} \times \frac{50}{60} + \left(\frac{9 - 1}{9 + 1} \right) \left(0.23 + \frac{0.12}{9} \right) = 2.06, \\ s &= \frac{w}{h} = \frac{8e^p}{e^{2p} - 2} = \frac{8e^{2.06}}{e^{4.12} - 2} = 1.056. \end{aligned}$$

Hence,

$$w = sh = 1.056 \times 0.5 \, \text{mm} = 0.53 \, \text{mm}.$$

To check our calculations, we use $s = 1.056$ to calculate Z_0 to verify that the value we obtained is indeed equal or close to $50 \, \Omega$. With $\epsilon_r = 9$, Eqs. (2.37a) to (2.40) yield

$$\begin{aligned} x &= 0.55, & y &= 0.99, \\ t &= 12.51, & \epsilon_{\text{eff}} &= 6.11, \\ Z_0 &= 49.93 \, \Omega. \end{aligned}$$

The calculated value of Z_0 is, for all practical purposes, equal to the value specified in the problem statement.

Exercise 2-5: A microstrip transmission line uses a strip of width w and height $h = 1 \, \text{mm}$ over a substrate of relative permittivity $\epsilon_r = 4$. What should w be so that the characteristic impedance of the line is $Z_0 = 50 \, \Omega$?

Answer: $w = 2.05 \, \text{mm}$. (See [EM](#).)

2-6 The Lossless Transmission Line: General Considerations

According to the preceding section, a transmission line is fully characterized by two fundamental parameters: its propagation constant γ and its characteristic impedance Z_0 , both of which are specified by the angular frequency ω and the line parameters R' , L' , G' , and C' .

► In many practical situations, the transmission line can be designed to exhibit low ohmic losses by selecting conductors with very high conductivities and dielectric materials (separating the conductors) with negligible conductivities. As a result, R' and G' assume very small values such that $R' \ll \omega L'$ and $G' \ll \omega C'$. ◀

These conditions allow us to set $R' = G' \approx 0$ in Eq. (2.22), which yields

$$\gamma = \alpha + j\beta = j\omega\sqrt{L'C'}, \quad (2.44)$$

which in turn implies that

$$\alpha = 0 \quad \textbf{(lossless line)}, \quad (2.45a)$$

$$\beta = \omega\sqrt{L'C'} \quad \textbf{(lossless line)}. \quad (2.45b)$$

s 为 Z_0 的指定值。图形方法的逻辑扩展是生成提供 s 高精度估计的曲线拟合表达式。下列公式的误差小于2%：

(a) 对于 $Z_0 \leq (44 - 2\epsilon_r) \, \Omega$,

$$s = \frac{w}{h} = \frac{2}{\pi} \left\{ (q - 1) - \ln(2q - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left[\ln(q - 1) + 0.29 - \frac{0.52}{\epsilon_r} \right] \right\} \quad (2.42)$$

和

$$q = \frac{60\pi^2}{Z_0\sqrt{\epsilon_r}},$$

和

(b) 对于 $Z_0 \geq (44 - 2\epsilon_r) \, \Omega$,

$$s = \frac{w}{h} = \frac{8e^p}{e^{2p} - 2}, \quad (2.43a)$$

和

$$p = \sqrt{\frac{\epsilon_r + 1}{2}} \frac{Z_0}{60} + \left(\frac{\epsilon_r - 1}{\epsilon_r + 1} \right) \left(0.23 + \frac{0.12}{\epsilon_r} \right). \quad (2.43b)$$

上述表达式假定介电基板的相对介电常数 ϵ_r 已被指定。对于包括 Duroid、Teflon、硅和蓝宝石在内的典型基板材料， ϵ_r 范围在 2 到 15 之间。

例 2-2：微带线

50Ω微带线采用0.5mm厚的蓝宝石基板， $\epsilon_r = 9$.其铜带宽度是多少？

解：由于 $Z_0 = 50 > 44 - 18 = 32$ ，我们应该使用方程：(2.43)：

$$\begin{aligned} p &= \sqrt{\frac{\epsilon_r + 1}{2}} \times \frac{Z_0}{60} + \left(\frac{\epsilon_r - 1}{\epsilon_r + 1} \right) \left(0.23 + \frac{0.12}{\epsilon_r} \right) \\ &= \sqrt{\frac{9 + 1}{2}} \times \frac{50}{60} + \left(\frac{9 - 1}{9 + 1} \right) \left(0.23 + \frac{0.12}{9} \right) = 2.06, \\ s &= \frac{w}{h} = \frac{8e^p}{e^{2p} - 2} = \frac{8e^{2.06}}{e^{4.12} - 2} = 1.056. \end{aligned}$$

因此，

$$w = sh = 1.056 \times 0.5 \, \text{mm} = 0.53 \, \text{mm}.$$

为了检查我们的计算，我们使用 $s = 1.056$ 来计算 Z_0 ，以验证我们获得的值确实等于或接近 $50 \, \Omega$ 。对于 $\epsilon_r = 9$ ，等式。 (2.37a) 至 (2.40) 产率

$$\begin{aligned} x &= 0.55, & y &= 0.99, \\ t &= 12.51, & \epsilon_{\text{eff}} &= 6.11, \\ Z_0 &= 49.93 \, \Omega. \end{aligned}$$

出于所有实际目的， Z_0 的计算值等于问题陈述中指定的值。

练习 2-5：微带传输线在相对介电常数为 $\epsilon_r =$ 的基板上使用宽度为 w 、高度为 $h = 1 \, \text{mm}$ 的条带 4. 为了使线路的特性阻抗为 $Z_0 = 50 \, \Omega$ ， w 应该是多少？

答案： $w = 2.05$ 毫米。（参见^{EM}。）

2-6 无损传输线：一般考虑因素

根据上一节，传输线完全由两个基本参数表征：传播常数 γ 和特性阻抗 Z_0 ，这两个参数均由角频率 ω 和线路参数 R' 、 L' 、 G' 和 C' 指定。

► 在许多实际情况下，通过选择电导率非常高的导体和电导率可忽略不计的介电材料（分隔导体），传输线可以设计成具有低欧姆损耗。结果， R' 和 G' 呈现非常小的值，使得 $R' \ll \omega L'$ 和 $G' \ll \omega C'$ 。 ◀

这些条件允许我们在方程 (1) 中设置 $R' = G' \approx 0$ 。 (2.22)，得出

$$\gamma = \alpha + j\beta = j\omega\sqrt{L'C'}, \quad (2.44)$$

这反过来意味着

$$\alpha = 0 \quad \textbf{(lossless line)}, \quad (2.45a)$$

$$\beta = \omega\sqrt{L'C'} \quad \textbf{(lossless line)}. \quad (2.45b)$$

For the characteristic impedance, application of the lossless line conditions to Eq. (2.29) leads to

$$Z_0 = \sqrt{\frac{L'}{C'}} \quad \textbf{(lossless line)}, \quad (2.46)$$

which now is a real number. Using the lossless line expression for β [Eq. (2.45b)], we obtain the following expressions for the guide wavelength λ and the phase velocity u_p :

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{LC'}}, \quad (2.47)$$

$$u_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC'}}. \quad (2.48)$$

Upon using Eq. (2.10), Eqs. (2.45b) and (2.48) may be rewritten as

$$\beta = \omega\sqrt{\mu\varepsilon} \quad (\text{rad/m}), \quad (2.49)$$

$$u_p = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\text{m/s}), \quad (2.50)$$

where μ and ε are, respectively, the magnetic permeability and electrical permittivity of the insulating material separating the conductors. Materials used for this purpose are usually characterized by a permeability $\mu_0 = 4\pi \times 10^{-7}$ H/m (the permeability of free space). Also, the permittivity ε is often specified in terms of the relative permittivity ε_r defined as

$$\varepsilon_r = \varepsilon/\varepsilon_0, \quad (2.51)$$

where $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m $\approx (1/36\pi) \times 10^{-9}$ F/m is the permittivity of free space (vacuum). Hence, Eq. (2.50) becomes

$$u_p = \frac{1}{\sqrt{\mu_0\varepsilon_r\varepsilon_0}} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \cdot \frac{1}{\sqrt{\varepsilon_r}} = \frac{c}{\sqrt{\varepsilon_r}}, \quad (2.52)$$

where $c = 1/\sqrt{\mu_0\varepsilon_0} = 3 \times 10^8$ m/s is the velocity of light in free space. If the insulating material between the conductors is air, then $\varepsilon_r = 1$ and $u_p = c$. In view of Eq. (2.51) and the relationship between λ and u_p given by Eq. (2.33), the wavelength is given by

$$\lambda = \frac{u_p}{f} = \frac{c}{f} \frac{1}{\sqrt{\varepsilon_r}} = \frac{\lambda_0}{\sqrt{\varepsilon_r}}, \quad (2.53)$$

where $\lambda_0 = c/f$ is the wavelength in air corresponding to a frequency f . Note that, because both u_p and λ depend on ε_r , the

choice of the type of insulating material used in a transmission line is dictated not only by its mechanical properties but by its electrical properties as well.

According to Eq. (2.52), if ε_r of the insulating material is independent of f (which usually is the case for commonly used TEM lines), the same independence applies to u_p .

► If sinusoidal waves of different frequencies travel on a transmission line with the same phase velocity, the line is called **nondispersive**. ◀

This is an important feature to consider when digital data are transmitted in the form of pulses. A rectangular pulse or a series of pulses is composed of many Fourier components with different frequencies. If the phase velocity is the same for all frequency components (or at least for the dominant ones), then the pulse's shape does not change as it travels down the line. In contrast, the shape of a pulse propagating in a dispersive medium becomes progressively distorted, and the pulse length increases (stretches out) as a function of the distance traveled in the medium (**Fig. 2-3**), thereby imposing a limitation on the maximum data rate (which is related to the length of the individual pulses and the spacing between adjacent pulses) that can be transmitted through the medium without loss of information.

Table 2-2 provides a list of the expressions for γ , Z_0 , and u_p for the general case of a lossy line and for several types of lossless lines. The expressions for the lossless lines are based on the equations for L' and C' given in **Table 2-1**.

Exercise 2-6: For a lossless transmission line, $\lambda = 20.7$ cm at 1 GHz. Find ε_r of the insulating material.

Answer: $\varepsilon_r = 2.1$. (See [EM](#).)

Exercise 2-7: A lossless transmission line uses a dielectric insulating material with $\varepsilon_r = 4$. If its line capacitance is $C' = 10$ (pF/m), find (a) the phase velocity u_p , (b) the line inductance L' , and (c) the characteristic impedance Z_0 .

Answer: (a) $u_p = 1.5 \times 10^8$ (m/s), (b) $L' = 4.45$ (μ H/m), (c) $Z_0 = 667.1$ Ω . (See [EM](#).)

2-6.1 Voltage Reflection Coefficient

With $\gamma = j\beta$ for the lossless line, Eqs. (2.26a) and (2.30) for the total voltage and current become

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \quad (2.54a)$$

对于特性阻抗，将无损线路条件应用到等式：(2.29) 导致

$$Z_0 = \sqrt{\frac{L'}{C'}} \quad \textbf{(lossless line)}, \quad (2.46)$$

现在是一个实数。使用 β 的无损线表达式 [Eq. (2.45b)], 我们得到波导波长 λ 和相速度 u_p 的表达式：

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{LC'}}, \quad (2.47)$$

$$u_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC'}}. \quad (2.48)$$

使用方程式后。(2.10) , 等式。(2.45b) 和 (2.48) 可以重写为

$$\beta = \omega\sqrt{\mu\varepsilon} \quad (\text{rad/m}), \quad (2.49)$$

$$u_p = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\text{m/s}), \quad (2.50)$$

其中 μ 和 ε 分别是分隔导体的绝缘材料的磁导率和介电常数。用于此目的的材料通常以磁导率 $\mu_0 = 4\pi \times 10^{-7}$ H/m (自由空间的磁导率) 为特征。此外, 介电常数 ε 通常用相对介电常数 ε_r 来指定, 定义为

$$\varepsilon_r = \varepsilon/\varepsilon_0, \quad (2.51)$$

其中 $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m $\approx (1/36\pi) \times 10^{-9}$ F/m 是自由空间 (真空) 的介电常数。因此, 等式。(2.50) 变为

$$u_p = \frac{1}{\sqrt{\mu_0\varepsilon_r\varepsilon_0}} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \cdot \frac{1}{\sqrt{\varepsilon_r}} = \frac{c}{\sqrt{\varepsilon_r}}, \quad (2.52)$$

其中 $c = 1/\sqrt{\mu_0\varepsilon_0} = 3 \times 10^8$ m/s 是自由空间中的光速。如果导体之间的绝缘材料是空气, 则 $\varepsilon_r = 1$ 和 $u_p = c$ 。鉴于方程。(2.51) 以及方程 (2.51) 给出的 λ 和 u_p 之间的关系(2.33), 波长由下式给出

$$\lambda = \frac{u_p}{f} = \frac{c}{f} \frac{1}{\sqrt{\varepsilon_r}} = \frac{\lambda_0}{\sqrt{\varepsilon_r}}, \quad (2.53)$$

其中 $\lambda_0 = c/f$ 是空气中对应于频率 f 的波长。请注意, 由于 u_p 和 λ 都依赖于 ε_r , 因此

传输线中使用的绝缘材料类型的选择不仅取决于其机械性能, 还取决于其电气性能。

根据方程。根据方程 (2.52), 如果绝缘材料的 ε_r 与 f 无关 (对于常用的 TEM 线通常就是这种情况) , 则同样的独立性也适用于 u_p 。

► 如果不同频率的正弦波以相同的相速度在传输线上传播, 则该传输线称为非色散传输线。 ◀

当数字数据以脉冲形式传输时, 这是一个需要考虑的重要特征。矩形脉冲或一系列脉冲由许多不同频率的傅立叶分量组成。如果所有频率分量 (或至少主要频率分量) 的相速度相同, 则脉冲的形状在沿线路传播时不会改变。相比之下, 在色散介质中传播的脉冲形状逐渐扭曲, 并且脉冲长度随着在介质中传播的距离而增加 (拉长) (图2-3), 从而对可以通过介质传输而不丢失信息的最大数据速率 (与单个脉冲的长度和相邻脉冲之间的间距有关) 施加限制。

表 2-2 提供了有损线路的一般情况和几种类型的无损线路的 γ 、 Z_0 和 u_p 的表达式列表。无损线路的表达式基于表 2-1 中给出的 L' 和 C' 方程。

练习 2-6: 对于无损传输线, 1 GHz 时 $\lambda = 20.7$ cm。找到绝缘材料的 ε_r 。

Answer: $\varepsilon_r = 2.1$. (See [EM](#).)

练习 2-7: 无损传输线使用 $\varepsilon_r =$ 的电介质绝缘材料 4 。如果其线路电容为 $C' = 10$ (pF/m), 则求 (a) 相速度 u_p , (b) 线路电感 L' , 以及 (c) 特性阻抗 Z_0 。

答案: (a) $u_p = 1.5 \times 10^8$ (m/s), (b) $L' = 4.45$ (μ H/m) , (c) $Z_0 = 667.1$ Ω 。 ([参见EM](#)。)

2-6.1 电压反射系数

对于无损线, 方程为 $\gamma = j\beta$ 。(2.26a) 和 (2.30) 的总电压和电流变为

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \quad (2.54a)$$

Table 2-2 Characteristic parameters of transmission lines.

	Propagation Constant $\gamma = \alpha + j\beta$	Phase Velocity u_p	Characteristic Impedance Z_0
General case	$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}$	$u_p = \omega / \beta$	$Z_0 = \sqrt{\frac{(R' + j\omega L')}{(G' + j\omega C')}}$
Lossless ($R' = G' = 0$)	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = \sqrt{L' / C'}$
Lossless coaxial	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = (60 / \sqrt{\epsilon_r}) \ln(b / a)$
Lossless two-wire	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = (120 / \sqrt{\epsilon_r})$ $\cdot \ln \left[(D / d) + \sqrt{(D / d)^2 - 1} \right]$ $Z_0 \approx (120 / \sqrt{\epsilon_r}) \ln(2D / d),$ if $D \gg d$
Lossless parallel-plate	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = (120\pi / \sqrt{\epsilon_r})(h / w)$
Notes: (1) $\mu = \mu_0$, $\epsilon = \epsilon_r \epsilon_0$, $c = 1 / \sqrt{\mu_0 \epsilon_0}$, and $\sqrt{\mu_0 / \epsilon_0} \approx (120\pi) \Omega$, where ϵ_r is the relative permittivity of insulating material. (2) For coaxial line, a and b are radii of inner and outer conductors. (3) For two-wire line, d = wire diameter and D = separation between wire centers. (4) For parallel-plate line, w = width of plate and h = separation between the plates.			

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}. \quad (2.54b)$$

These expressions contain two unknowns, V_0^+ and V_0^- . According to Section 1-7.2, an exponential factor of the form $e^{-j\beta z}$ is associated with a wave traveling in the positive z direction, from the source (sending end) to the load (receiving end). Accordingly, we refer to it as the *incident wave* with V_0^+ as its voltage amplitude. Similarly, the term containing $V_0^- e^{j\beta z}$ represents a *reflected wave* with voltage amplitude V_0^- traveling along the negative z direction from the load to the source.

To determine V_0^+ and V_0^- , we need to consider the lossless transmission line in the context of the complete circuit, including a generator circuit at its input terminals and a load at its output terminals, as shown in Fig. 2-12. The line of length l is terminated in an arbitrary *load impedance* Z_L .

► For mathematical convenience, the reference of the spatial coordinate z is chosen such that $z = 0$ corresponds to the location of the load, not the generator. ◀

At the sending end, at $z = -l$, the line is connected to a sinusoidal voltage source with phasor voltage $\tilde{V}_g$ and internal impedance Z_g . Since z points from the generator to the load, positive values of z correspond to locations beyond the load; therefore, they are irrelevant to our circuit. In future sections, we will find it more convenient to work with a spatial dimension that also starts at the load but whose direction is opposite of z . We shall call it the *distance from the load* d and define it as $d = -z$, as shown in Fig. 2-12.

The phasor voltage across the load, $\tilde{V}_L$, and the phasor current through it, $\tilde{I}_L$, are related by the load impedance Z_L as

$$Z_L = \frac{\tilde{V}_L}{\tilde{I}_L}. \quad (2.55)$$

The voltage $\tilde{V}_L$ is the total voltage on the line $\tilde{V}(z)$ given by Eq. (2.54a), and $\tilde{I}_L$ is the total current $\tilde{I}(z)$ given by Eq. (2.54b). Both are evaluated at $z = 0$:

$$\tilde{V}_L = \tilde{V}(z=0) = V_0^+ + V_0^-, \quad (2.56a)$$

表2-2 传输线特性参数。

	Propagation Constant $\gamma = \alpha + j\beta$	Phase Velocity u_p	Characteristic Impedance Z_0
General case	$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}$	$u_p = \omega / \beta$	$Z_0 = \sqrt{\frac{(R' + j\omega L')}{(G' + j\omega C')}}$
Lossless ($R' = G' = 0$)	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = \sqrt{L' / C'}$
Lossless coaxial	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = (60 / \sqrt{\epsilon_r}) \ln(b / a)$
Lossless two-wire	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = (120 / \sqrt{\epsilon_r})$ $\cdot \ln \left[(D / d) + \sqrt{(D / d)^2 - 1} \right]$ $Z_0 \approx (120 / \sqrt{\epsilon_r}) \ln(2D / d),$ if $D \gg d$
Lossless parallel-plate	$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$	$u_p = c / \sqrt{\epsilon_r}$	$Z_0 = (120\pi / \sqrt{\epsilon_r})(h / w)$

Notes: (1) $\mu = \mu_0$, $\epsilon = \epsilon_r \epsilon_0$, $c = 1 / \sqrt{\mu_0 \epsilon_0}$, and $\sqrt{\mu_0 / \epsilon_0} \approx (120\pi) \Omega$, where ϵ_r is the relative permittivity of insulating material. (2) For coaxial line, a and b are radii of inner and outer conductors. (3) For two-wire line, d = wire diameter and D = separation between wire centers. (4) For parallel-plate line, w = width of plate and h = separation between the plates.

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}. \quad (2.54b)$$

这些表达式包含两个未知数 V_0^+ 和 V_0^- 。根据第 1-7.2 节， $e^{-j\beta z}$ 形式的指数因子与沿正 z 方向从源（发送端）到负载（接收端）传播的波相关。因此，我们将其称为以 V_0^+ 为其电压幅度的入射波。类似地，包含 $V_0^- e^{j\beta z}$ 的项表示电压幅度为 V_0^- 的反射波，沿负 z 方向从负载传播到源。

为了确定 V_0^+ 和 V_0^- ，我们需要考虑完整电路中的无损传输线，包括输入端的发生器电路和输出端的负载，如图 2-12 所示。长度为 l 的线路以任意负载阻抗 Z_L 终止。

► 为了数学方便，选择空间坐标 z 的参考，使得 $z = 0$ 对应于负载的位置，而不是发电机的位置。 ◀

在发送端，在 $z = -l$ 处，线路连接到具有相量电压 $\tilde{V}_g$ 和内部阻抗 Z_g 的正弦电压源。由于 z 点从发电机到负载，因此 z 的正值对应于超出负载的位置；因此，它们与我们的电路无关。在以后的章节中，我们会发现使用同样从载荷开始但方向与 z 相反的空间维度会更方便。我们将其称为距负载的距离 d ，并将其定义为 $d = -z$ ，如图 2-12 所示。

负载两端的相量电压 $\tilde{V}_L$ 和流过负载的相量电流 $\tilde{I}_L$ 与负载阻抗 Z_L 的关系如下：

$$Z_L = \frac{\tilde{V}_L}{\tilde{I}_L}. \quad (2.55)$$

电压 $\tilde{V}_L$ 是等式 1 给出的线上 $\tilde{V}(z)$ 上的总电压。(2.54a)， $\tilde{I}_L$ 是由式 (2.54a) 给出的总电流 $\tilde{I}(z)$ 。(2.54b)。两者均在 $z = 0$ 处求值：

$$\tilde{V}_L = \tilde{V}(z=0) = V_0^+ + V_0^-, \quad (2.56a)$$

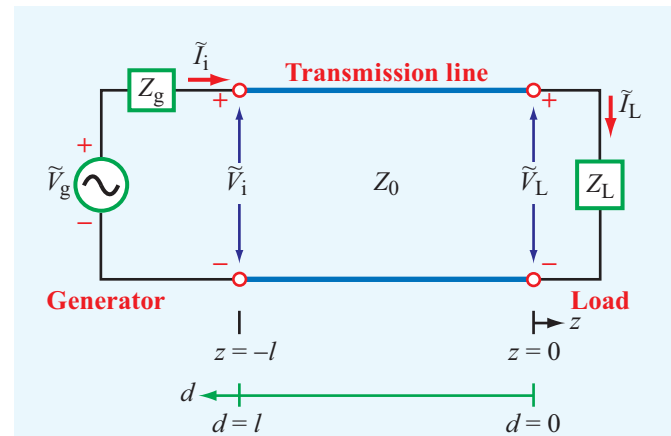


Figure 2-12 Transmission line of length l connected on one end to a generator circuit and on the other end to a load Z_L . The load is located at $z = 0$, and the generator terminals are at $z = -l$. Coordinate d is defined as $d = -z$.

is the **normalized load impedance**. In many transmission line problems, we can streamline the necessary computation by normalizing all impedances in the circuit to the characteristic impedance Z_0 . *Normalized impedances are denoted by lower-case letters.*

In view of Eq. (2.28), the ratio of the current amplitudes is

$$\frac{I_0^-}{I_0^+} = -\frac{V_0^-}{V_0^+} = -\Gamma. \quad (2.61)$$

► We note that whereas the ratio of the voltage amplitudes is equal to Γ , the ratio of the current amplitudes is equal to $-\Gamma$. ◀

The reflection coefficient Γ is governed by a single parameter, the normalized load impedance z_L . As indicated by Eq. (2.46), Z_0 of a lossless line is a real number. However, Z_L is in general a complex quantity, as in the case of a series RL circuit, for example, for which $Z_L = R + j\omega L$. Hence, in general Γ also is complex and given by

$$\Gamma = |\Gamma|e^{j\theta_r}, \quad (2.62)$$

where $|\Gamma|$ is the magnitude of Γ and θ_r is its phase angle. Note that $|\Gamma| \leq 1$.

► A load is said to be **matched** to a transmission line if $Z_L = Z_0$ because then there will be no reflection by the load ($\Gamma = 0$ and $V_0^- = 0$). ◀

On the other hand, when the load is an open circuit ($Z_L = \infty$), $\Gamma = 1$ and $V_0^- = V_0^+$, and when it is a short circuit ($Z_L = 0$), $\Gamma = -1$ and $V_0^- = -V_0^+$ (Table 2-3).

Example 2-3: Reflection Coefficient of a Series RC Load

A $100 \, \Omega$ transmission line is connected to a load consisting of a $50 \, \Omega$ resistor in series with a $10 \, \text{pF}$ capacitor. Find the reflection coefficient at the load for a $100 \, \text{MHz}$ signal.

Solution: The following quantities are given (Fig. 2-13):

$$\begin{aligned} R_L &= 50 \, \Omega, & C_L &= 10 \, \text{pF} = 10^{-11} \, \text{F}, \\ Z_0 &= 100 \, \Omega, & f &= 100 \, \text{MHz} = 10^8 \, \text{Hz}. \end{aligned}$$

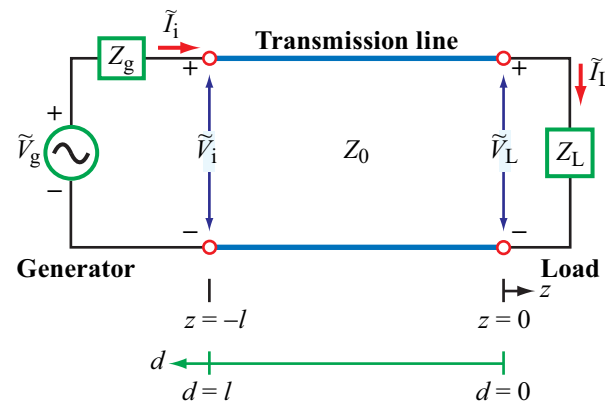


图 2-12 长度为 l 的传输线，一端连接到发电电路，另一端连接到负载 Z_L 。负载位于 $z=0$ ，发电机端子位于 $z=-l$ 。坐标 d 定义为 $d = -z$ 。

是归一化负载阻抗。在许多传输线问题中，我们可以通过将电路中的所有阻抗归一化为特性阻抗 Z_0 来简化必要的计算。归一化阻抗用小写字母表示。

鉴于方程。 (2.28) 中，电流幅值之比为

$$\frac{I_0^-}{I_0^+} = -\frac{V_0^-}{V_0^+} = -\Gamma. \quad (2.61)$$

► 我们注意到，电压幅值之比等于 Γ ，而电流幅值之比等于 $-\Gamma$ 。 ◀

反射系数 Γ 由单个参数（归一化负载阻抗 z_L ）控制。如方程所示。 (2.46)，无损线的 Z_0 是实数。然而， Z_L 通常是复数，例如在串联 RL 电路的情况下，其中 $Z_L = R + j\omega L$ 。因此，一般来说 Γ 也是复数，由下式给出

$$\tilde{I}_L = \tilde{I}(z=0) = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}. \quad (2.56b)$$

在等式中使用这些表达式。 (2.55)，我们得到

$$Z_L = \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} \right) Z_0. \quad (2.57)$$

求解 V_0^- 给出

$$V_0^- = \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) V_0^+. \quad (2.58)$$

► 负载处反射电压波和入射电压波的振幅之比称为电压反射系数 Γ 。 ◀

$$\Gamma = |\Gamma|e^{j\theta_r}, \quad (2.62)$$

其中 $|\Gamma|$ 是 Γ 的大小， θ_r 是其相位角。请注意 $|\Gamma| \leq 1$ 。

► 如果 $Z_L = Z_0$ ，则称负载与传输线匹配，因为负载不会产生反射 ($\Gamma = 0$ 和 $V_0^- = 0$)。 ◀

另一方面，当负载开路 ($Z_L = \infty$) 时， $\Gamma = 1$ 和 $V_0^- = V_0^+$ ，当负载短路 ($Z_L = 0$) 时， $\Gamma = -1$ 和 $V_0^- = -V_0^+$ (表 2-3)。

示例 2-3：串联 RC 负载的反射系数

$100 \, \Omega$ 传输线连接到由 $50 \, \Omega$ 电阻器与 $10 \, \text{pF}$ 电容器串联组成的负载。求 $100 \, \text{MHz}$ 信号负载处的反射系数。

解：给出下列数量 (图 2-13)：

$$\begin{aligned} R_L &= 50 \, \Omega, & C_L &= 10 \, \text{pF} = 10^{-11} \, \text{F}, \\ Z_0 &= 100 \, \Omega, & f &= 100 \, \text{MHz} = 10^8 \, \text{Hz}. \end{aligned}$$

$$\tilde{I}_L = \tilde{I}(z=0) = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}. \quad (2.56b)$$

Using these expressions in Eq. (2.55), we obtain

$$Z_L = \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} \right) Z_0. \quad (2.57)$$

Solving for V_0^- gives

$$V_0^- = \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) V_0^+. \quad (2.58)$$

► The ratio of the amplitudes of the reflected and incident voltage waves at the load is known as the **voltage reflection coefficient** Γ . ◀

From Eq. (2.58), it follows that

$$\begin{aligned} \Gamma &= \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} \\ &= \frac{z_L - 1}{z_L + 1} \quad (\text{dimensionless}), \end{aligned} \quad (2.59)$$

where

$$z_L = \frac{Z_L}{Z_0} \quad (2.60)$$

在哪里

$$z_L = \frac{Z_L}{Z_0} \quad (2.60)$$